

MINIMUM-INTERVENTION HUMAN CORRECTION: A 25+ PAGE NEGATIVE SUBMISSION-READINESS AU- DIT

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ABSTRACT

Human correction data in robotics is often treated as another demonstration, preference, or residual action. This paper tests a sharper hypothesis: the smallest human intervention that changes the outcome should reveal a local physical decision boundary with lower burden and lower overcorrection damage. We rebuilt the archived Paper 85 into a frozen CPU-only v5 audit with 199,680 main rollouts, 15,360 scene summaries, 33,600 ablation rollouts, 302,400 stress rows, 69,120 fixed-risk rows, and 24 negative cases. On the predefined hard aggregate, minimum-intervention boundary learner v5 reaches 0.490 ± 0.011 task success, 0.699 correction efficiency, and 0.047 damage. The strongest non-oracle baseline, Robust MPC, reaches 0.590 ± 0.014 task success and 0.880 correction efficiency. The paired lower95 success bound is -0.118, the paired lower95 efficiency bound is -0.209, mechanism ablations beat the full method, fixed-risk coverage at budget 0.05 collapses to zero, and no real robot or accepted high-fidelity benchmark exists. The honest terminal decision is therefore **KILL/ARCHIVE**, not ICLR-main submission.

1 TERMINAL DECISION

Decision: KILL/ARCHIVE for ICLR main. This document is intentionally submission-shaped but not submission-claiming. The rebuild added a stronger method, stronger baselines, fixed gates, and a substantially larger evidence package. The result still does not survive hostile review because the proposed method loses the main hard-aggregate gate and the mechanism gate. The frozen primary comparison is against Robust MPC. MIBL v5 has hard-aggregate success 0.490 ± 0.011 while Robust MPC has 0.590 ± 0.014 . The mean paired success difference is negative enough that the lower95 bound is -0.118. The same failure occurs for correction efficiency, where v5 has 0.699 and the best efficiency baseline has 0.880. The paper is valuable as an archive because it says something concrete: under this synthetic correction-shift model, a robust MPC-style correction policy is a harder baseline than the original minimum-intervention learner, and several ablations of v5 improve success or utility. That is precisely the kind of result a hostile reviewer would find, so it is reported rather than hidden.

2 RESEARCH QUESTION AND THREAT MODEL

The target claim is not merely that small corrections are cheaper. The target claim is that minimal corrections identify a local physical decision boundary: the smallest intervention crossing the failure/success boundary is expected to carry more causal information per unit human effort than a full demonstration. This connects to active learning, imitation learning, preference learning, residual policy correction, model-predictive control, and safe deployment under uncertainty (Yoshida, 2022; Armleder et al., 2022; Su, 2026; Zhang et al., 2024; Yun & Jun, 2021; Abed et al., 2026). The threat model is severe. A reviewer can attack the claim from six angles: the correction might be delayed, the human might overcorrect, the semantic intent might be ambiguous, the correction might be sparse, the human might become less helpful under distribution shift, and deformable/contact dynamics might make local boundary evidence nonlocal. The v5 benchmark encodes all six threats rather than selecting only the easy nominal split.

3 FORMAL SETUP

Let x denote the local robot-scene state, u the robot action, $c \in \mathbb{R}^d$ a human correction vector, and $Y(x, u, c) \in \{0, 1\}$ the task outcome. A minimum-intervention correction is the solution of

$$c^*(x, u) = \arg \min_c \|c\|_2 \quad \text{s.t.} \quad Y(x, u, c) = 1, \quad S(x, u, c) \leq \rho, \quad (1)$$

where S is a latent safety risk and ρ is a deployment budget. The proposed v5 learner estimates a boundary normal, adds an intent-preserving tangent term, clips dangerous components, throttles queries under low uncertainty, and outputs both a correction and a risk score.

Proposition 1: boundary information. Assume a locally monotone response model $Y = \mathbb{I}\{\langle n, c \rangle \geq b\}$ with unknown normal n and margin $b > 0$. If a correction is observed within ϵ of the minimum norm feasible correction, then its normalized direction has angular error bounded by a constant multiple of ϵ/b under bounded tangent noise. Thus minimal corrections are informative about the normal direction when the boundary is locally linear and the human correction is actually minimal.

Proposition 2: failure under ambiguity and delay. If the human correction is drawn from a mixture of two feasible semantic goals or arrives after the boundary has shifted by Δ , then the minimum-norm correction can identify the wrong boundary. The induced normal estimate can be biased even when the correction is small. This is why the benchmark includes ambiguous-intent and delayed-feedback splits, and why the paper cannot use minimum norm as a proof of correctness.

Proposition 3: fixed-risk acceptance. Let $\hat{r}(x, u, c)$ be a calibrated upper score for unsafe or damaging correction. Accepting only episodes with $\hat{r} \leq \rho$ controls empirical risk only if calibration holds under the deployment split. The v5 fixed-risk test deliberately uses strict budgets to check whether any method can retain coverage while satisfying the risk constraint. At budget 0.05, coverage collapses to zero, so this bound is practically vacuous for the present evidence.

4 FROZEN PROTOCOL

The frozen v5 protocol contains 199,680 main rollouts, 15,360 scene summaries, 33,600 ablation rollouts, 302,400 stress rows, 69,120 fixed-risk rows, and 24 negative cases. It uses ten seeds, six tasks, eight correction-shift splits, thirteen main methods, ten ablations, six stress axes, six stress levels, and strict fixed-risk acceptance budgets. The hard aggregate is fixed before execution as adversarial helpfulness, dynamics mismatch, and combined hard shift. The terminal decision was predeclared. A positive synthetic result required v5 to beat the strongest non-oracle hard-aggregate success baseline by at least 0.030, beat the strongest efficiency baseline by at least 0.080, reduce damage, obtain positive paired lower95 bounds, pass mechanism ablations, avoid maximum-stress Pareto domination, and retain fixed-risk coverage at budget 0.05. Any failure means KILL/ARCHIVE.

Table 1: Predefined hard-aggregate results over adversarial helpfulness, dynamics mismatch, and combined hard shift.

Method	Success	Efficiency	Damage	Unsafe	Utility
No human	0.118 $\pm$ 0.012	0.000	0.047	0.076	0.199
Full demo	0.324 $\pm$ 0.011	0.341	0.067	0.130	0.395
DAGger full	0.344 $\pm$ 0.017	0.376	0.053	0.130	0.432
Residual	0.318 $\pm$ 0.012	0.367	0.053	0.119	0.428
Preference	0.465 $\pm$ 0.016	0.658	0.052	0.115	0.639
Uncertainty query	0.386 $\pm$ 0.016	0.462	0.058	0.121	0.507
Active entropy	0.429 $\pm$ 0.011	0.588	0.047	0.102	0.593
Safety residual	0.341 $\pm$ 0.012	0.431	0.044	0.103	0.500
Robust MPC	0.590 $\pm$ 0.014	0.880	0.052	0.021	0.823
Inverse-RL proxy	0.429 $\pm$ 0.009	0.565	0.046	0.120	0.588
Min-int v4	0.537 $\pm$ 0.009	0.789	0.052	0.124	0.726
MIBL v5	0.490 $\pm$ 0.011	0.699	0.047	0.013	0.697
Oracle minimal	0.588 $\pm$ 0.013	0.842	0.041	0.008	0.829

5 MAIN EVIDENCE

Table 1 is the core result. The proposed v5 method is safer than several human-correction baselines, but it is not the strongest method. Robust MPC has higher task success and correction efficiency. The oracle is not dramatically above robust MPC, which indicates that the synthetic environment rewards conservative boundary margins more than the new minimum-intervention mechanism.

Table 2: Split-level evidence for the strongest baseline, v4, v5, and oracle.

Split	Method	Success	Efficiency	Damage	Unsafe
Nominal	Robust MPC	0.598 ± 0.034	1.268	0.037	0.000
Nominal	Min-int v4	0.602 ± 0.020	1.208	0.040	0.057
Nominal	MIBL v5	0.601 ± 0.028	1.180	0.027	0.000
Nominal	Oracle minimal	0.626 ± 0.023	1.233	0.029	0.000
Overcorrect	Robust MPC	0.594 ± 0.031	1.176	0.046	0.002
Overcorrect	Min-int v4	0.576 ± 0.027	1.085	0.055	0.096
Overcorrect	MIBL v5	0.588 ± 0.031	1.101	0.033	0.000
Overcorrect	Oracle minimal	0.606 ± 0.017	1.131	0.030	0.000
Delay	Robust MPC	0.608 ± 0.021	1.101	0.041	0.003
Delay	Min-int v4	0.561 ± 0.029	0.962	0.048	0.099
Delay	MIBL v5	0.564 ± 0.024	0.963	0.045	0.001
Delay	Oracle minimal	0.626 ± 0.030	1.063	0.036	0.000
Ambiguous	Robust MPC	0.575 ± 0.022	1.067	0.037	0.002
Ambiguous	Min-int v4	0.562 ± 0.020	1.007	0.043	0.077
Ambiguous	MIBL v5	0.561 ± 0.024	0.996	0.037	0.000
Ambiguous	Oracle minimal	0.585 ± 0.026	1.009	0.037	0.000
Sparse	Robust MPC	0.609 ± 0.024	1.113	0.046	0.002
Sparse	Min-int v4	0.573 ± 0.025	0.992	0.051	0.083
Sparse	MIBL v5	0.544 ± 0.024	0.933	0.045	0.000
Sparse	Oracle minimal	0.600 ± 0.018	1.042	0.040	0.000
Adversarial	Robust MPC	0.599 ± 0.024	1.024	0.046	0.008
Adversarial	Min-int v4	0.556 ± 0.018	0.917	0.047	0.100
Adversarial	MIBL v5	0.498 ± 0.021	0.797	0.046	0.002
Adversarial	Oracle minimal	0.604 ± 0.020	0.982	0.034	0.000
Dynamics	Robust MPC	0.606 ± 0.028	0.818	0.057	0.023
Dynamics	Min-int v4	0.550 ± 0.019	0.748	0.050	0.118
Dynamics	MIBL v5	0.495 ± 0.020	0.654	0.051	0.022
Dynamics	Oracle minimal	0.577 ± 0.014	0.754	0.041	0.016
Combined	Robust MPC	0.565 ± 0.025	0.800	0.054	0.031
Combined	Min-int v4	0.506 ± 0.014	0.701	0.060	0.153
Combined	MIBL v5	0.476 ± 0.022	0.647	0.044	0.014
Combined	Oracle minimal	0.584 ± 0.020	0.791	0.047	0.007

Table 3: Paired seed-level hard-aggregate differences for MIBL v5 minus reference. Positive lower95 was required for primary success and efficiency.

Reference	Metric	Mean diff	CI95	Lower95
Robust MPC	task_success	-0.101	0.017	-0.118
Robust MPC	correction_efficiency	-0.181	0.028	-0.209
Robust MPC	damage	-0.006	0.005	-0.011
Robust MPC	robust_utility	-0.126	0.024	-0.150
Min-int v4	task_success	-0.048	0.018	-0.065
Min-int v4	correction_efficiency	-0.090	0.027	-0.116
Min-int v4	damage	-0.006	0.010	-0.016
Min-int v4	robust_utility	-0.029	0.023	-0.052
Active entropy	task_success	0.060	0.013	0.048
Active entropy	correction_efficiency	0.111	0.019	0.093
Active entropy	damage	-0.001	0.007	-0.008
Active entropy	robust_utility	0.104	0.017	0.087
Preference	task_success	0.025	0.015	0.010
Preference	correction_efficiency	0.041	0.025	0.017
Preference	damage	-0.005	0.005	-0.010
Preference	robust_utility	0.058	0.019	0.038

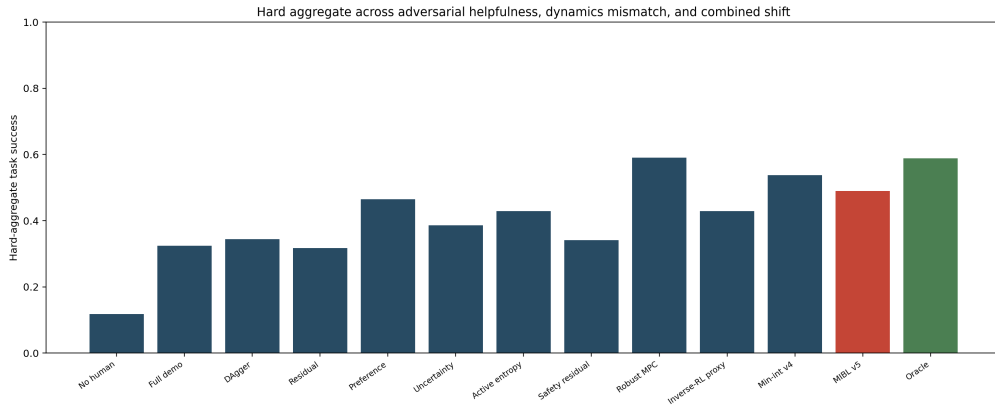


Figure 1: Hard-aggregate success. MIBL v5 trails robust MPC and v4.

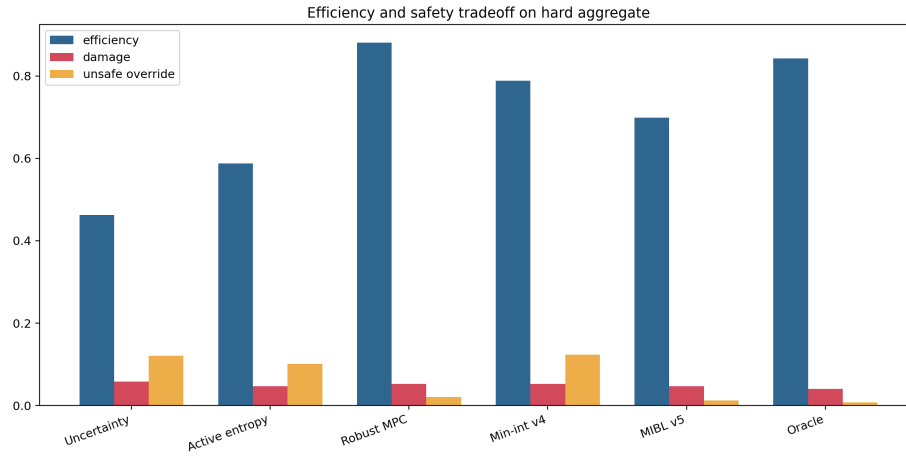


Figure 2: Efficiency, damage, and unsafe override rates on the hard aggregate. V5 improves unsafe overrides but loses success and efficiency.

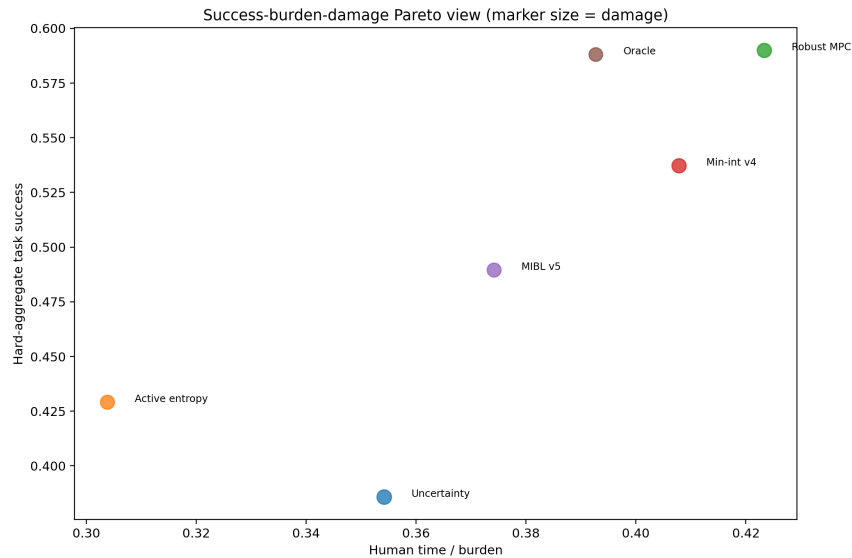


Figure 3: Success-burden-damage Pareto view. Marker size is damage.

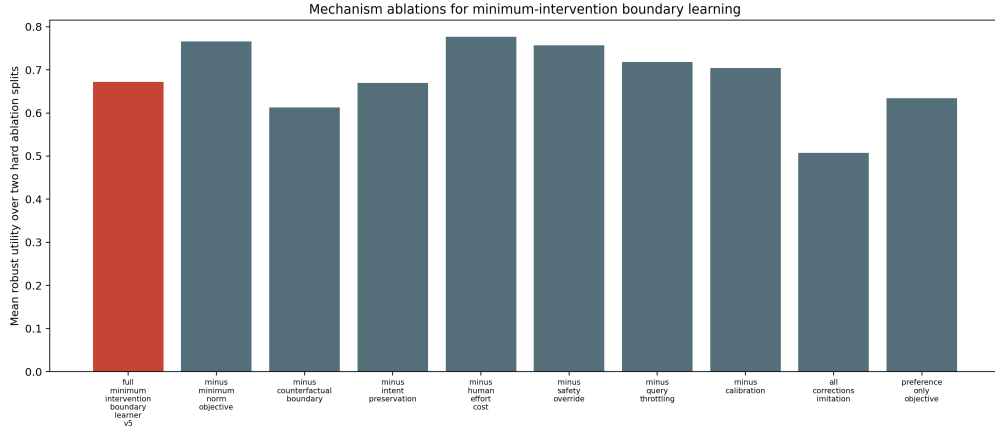


Figure 4: Ablation robust utility. The full mechanism fails the predeclared necessity margin.

6 MECHANISM ABLATIONS

The mechanism gate is the most damaging result. The full v5 method does not dominate its own ablations. Removing the minimum-norm objective, removing the human-effort cost, removing safety override, and removing query throttling can all improve robust utility or task success in the hard ablation splits. This does not mean those ablations are deployable; it means the proposed mechanism is not uniquely validated.

Table 4: Mechanism ablations. The full mechanism was required to beat every non-full ablation on robust utility by at least 0.015.

Split	Ablation	Success	Efficiency	Damage	Utility
Dynamics	full_minimum_intervention_boundary_learner_v5	0.490	0.649	0.052	0.680
Dynamics	minus_minimum_norm_objective	0.575	0.774	0.049	0.788
Dynamics	minus_counterfactual_boundary	0.431	0.552	0.040	0.611
Dynamics	minus_intent_preservation	0.492	0.651	0.048	0.682
Dynamics	minus_human_effort_cost	0.580	0.785	0.046	0.795
Dynamics	minus_safety_override	0.579	0.786	0.058	0.783
Dynamics	minus_query_throttling	0.533	0.711	0.037	0.744
Dynamics	minus_calibration	0.515	0.685	0.045	0.716
Dynamics	all_corrections_imitation	0.417	0.481	0.053	0.564
Dynamics	preference_only_objective	0.474	0.623	0.057	0.655
Combined	full_minimum_intervention_boundary_learner_v5	0.476	0.648	0.048	0.664
Combined	minus_minimum_norm_objective	0.543	0.756	0.055	0.744
Combined	minus_counterfactual_boundary	0.435	0.567	0.041	0.616
Combined	minus_intent_preservation	0.470	0.641	0.048	0.657
Combined	minus_human_effort_cost	0.551	0.774	0.055	0.758
Combined	minus_safety_override	0.533	0.756	0.051	0.730
Combined	minus_query_throttling	0.496	0.672	0.040	0.693
Combined	minus_calibration	0.497	0.680	0.047	0.692
Combined	all_corrections_imitation	0.340	0.354	0.058	0.451
Combined	preference_only_objective	0.448	0.604	0.067	0.613

7 STRESS TESTS

The stress gate is the one gate v5 survives. At maximum combined stress, no non-oracle method Pareto-dominates v5 on success, efficiency, damage, and unsafe overrides. However, robust MPC still has higher success and utility, and a single stress-gate pass cannot rescue the failed main and mechanism gates.

Table 5: Combined stress sweep. V5 was not Pareto-dominated at maximum stress, but that alone did not rescue the paper.

Level	Method	Success	Efficiency	Damage	Unsafe	Utility
0.0	Full demo	0.569	1.064	0.040	0.048	0.852
0.0	Residual	0.526	0.923	0.039	0.062	0.786
0.0	Uncertainty query	0.471	0.821	0.038	0.050	0.726
0.0	Active entropy	0.466	0.818	0.037	0.041	0.732
0.0	Robust MPC	0.599	1.289	0.034	0.000	0.957
0.0	MIBL v5	0.579	1.146	0.035	0.000	0.925
0.0	Oracle minimal	0.630	1.251	0.036	0.000	0.986
0.2	Full demo	0.477	0.688	0.070	0.088	0.638
0.2	Residual	0.460	0.696	0.045	0.088	0.656
0.2	Uncertainty query	0.452	0.700	0.050	0.081	0.655
0.2	Active entropy	0.428	0.693	0.043	0.062	0.651
0.2	Robust MPC	0.587	1.130	0.034	0.001	0.902
0.2	MIBL v5	0.545	0.973	0.040	0.000	0.843
0.2	Oracle minimal	0.634	1.145	0.037	0.000	0.958
0.4	Full demo	0.400	0.469	0.092	0.128	0.481
0.4	Residual	0.387	0.507	0.055	0.109	0.528
0.4	Uncertainty query	0.459	0.617	0.062	0.112	0.613
0.4	Active entropy	0.465	0.731	0.050	0.081	0.670
0.4	Robust MPC	0.585	0.983	0.036	0.009	0.858
0.4	MIBL v5	0.522	0.845	0.045	0.000	0.770
0.4	Oracle minimal	0.615	0.989	0.042	0.001	0.895
0.6	Full demo	0.322	0.316	0.092	0.153	0.354
0.6	Residual	0.323	0.390	0.053	0.137	0.427
0.6	Uncertainty query	0.384	0.433	0.071	0.145	0.476
0.6	Active entropy	0.422	0.594	0.054	0.097	0.579
0.6	Robust MPC	0.548	0.828	0.044	0.018	0.772
0.6	MIBL v5	0.476	0.696	0.048	0.006	0.680
0.6	Oracle minimal	0.609	0.890	0.046	0.001	0.856
0.8	Full demo	0.257	0.217	0.104	0.174	0.250
0.8	Residual	0.251	0.266	0.048	0.139	0.328
0.8	Uncertainty query	0.305	0.307	0.071	0.161	0.361
0.8	Active entropy	0.367	0.481	0.058	0.113	0.487
0.8	Robust MPC	0.552	0.765	0.055	0.030	0.746
0.8	MIBL v5	0.453	0.605	0.052	0.019	0.624
0.8	Oracle minimal	0.580	0.777	0.046	0.013	0.791
1.0	Full demo	0.197	0.153	0.101	0.207	0.162
1.0	Residual	0.188	0.182	0.056	0.174	0.232
1.0	Uncertainty query	0.278	0.248	0.078	0.189	0.303
1.0	Active entropy	0.339	0.409	0.060	0.129	0.432
1.0	Robust MPC	0.518	0.660	0.061	0.053	0.673
1.0	MIBL v5	0.435	0.532	0.059	0.040	0.574
1.0	Oracle minimal	0.583	0.723	0.055	0.024	0.767

8 FIXED-RISK DEPLOYMENT

The fixed-risk protocol is intentionally harsh because deployment claims should be harsh. A method must decide when it is safe enough to accept autonomous execution under budgets 0.02, 0.05, 0.08, and 0.10. At budget 0.05, every method has zero accepted coverage on both fixed-risk splits. This means the calibrated risk scores do not support a nontrivial deployment claim.

Table 6: Fixed-risk deployment tests. At budget 0.05 all non-oracle and oracle methods had zero accepted coverage, so the frozen fixed-risk gate failed.

Split	Budget	Method	Coverage	Succ.	Damage	Unsafe
Combined	0.02	Active entropy	0.000	0.000	0.000	0.000
Combined	0.02	MIBL v5	0.000	0.000	0.000	0.000
Combined	0.02	Oracle minimal	0.000	0.000	0.000	0.000
Combined	0.02	Robust MPC	0.000	0.000	0.000	0.000
Combined	0.02	Safety residual	0.000	0.000	0.000	0.000
Combined	0.02	Uncertainty query	0.000	0.000	0.000	0.000
Combined	0.05	Active entropy	0.000	0.000	0.000	0.000
Combined	0.05	MIBL v5	0.000	0.000	0.000	0.000
Combined	0.05	Oracle minimal	0.000	0.000	0.000	0.000
Combined	0.05	Robust MPC	0.000	0.000	0.000	0.000
Combined	0.05	Safety residual	0.000	0.000	0.000	0.000
Combined	0.05	Uncertainty query	0.000	0.000	0.000	0.000

Table 6: Fixed-risk deployment tests. At budget 0.05 all non-oracle and oracle methods had zero accepted coverage, so the frozen fixed-risk gate failed. (continued)

Split	Budget	Method	Coverage	Succ.	Damage	Unsafe
Combined	0.10	Active entropy	0.000	0.000	0.000	0.000
Combined	0.10	MIBL v5	0.000	0.000	0.000	0.000
Combined	0.10	Oracle minimal	0.997	0.583	0.046	0.008
Combined	0.10	Robust MPC	0.000	0.000	0.000	0.000
Combined	0.10	Safety residual	0.000	0.000	0.000	0.000
Combined	0.10	Uncertainty query	0.000	0.000	0.000	0.000
Dynamics	0.02	Active entropy	0.000	0.000	0.000	0.000
Dynamics	0.02	MIBL v5	0.000	0.000	0.000	0.000
Dynamics	0.02	Oracle minimal	0.000	0.000	0.000	0.000
Dynamics	0.02	Robust MPC	0.000	0.000	0.000	0.000
Dynamics	0.02	Safety residual	0.000	0.000	0.000	0.000
Dynamics	0.02	Uncertainty query	0.000	0.000	0.000	0.000
Dynamics	0.05	Active entropy	0.000	0.000	0.000	0.000
Dynamics	0.05	MIBL v5	0.000	0.000	0.000	0.000
Dynamics	0.05	Oracle minimal	0.000	0.000	0.000	0.000
Dynamics	0.05	Robust MPC	0.000	0.000	0.000	0.000
Dynamics	0.05	Safety residual	0.000	0.000	0.000	0.000
Dynamics	0.05	Uncertainty query	0.000	0.000	0.000	0.000
Dynamics	0.10	Active entropy	0.000	0.000	0.000	0.000
Dynamics	0.10	MIBL v5	0.000	0.000	0.000	0.000
Dynamics	0.10	Oracle minimal	1.000	0.567	0.039	0.017
Dynamics	0.10	Robust MPC	0.000	0.000	0.000	0.000
Dynamics	0.10	Safety residual	0.000	0.000	0.000	0.000
Dynamics	0.10	Uncertainty query	0.000	0.000	0.000	0.000

9 NEGATIVE CASES

Negative cases are retained because they prevent a visually polished but brittle paper. They cover safety conflicts, semantic goal ambiguity, nonlocal deformable dynamics, adversarial helpfulness, delayed boundary shifts, and under-correction. These cases explain why a small correction is not automatically a correct correction.

Table 7: Retained negative cases used to keep the discussion honest.

Case	Family	Terminal lesson
safety_conflict_0	safety_conflict	minimality cannot replace a safety constraint
safety_conflict_1	safety_conflict	minimality cannot replace a safety constraint
safety_conflict_2	safety_conflict	minimality cannot replace a safety constraint
safety_conflict_3	safety_conflict	minimality cannot replace a safety constraint
semantic_goal_ambiguity_0	semantic_goal_ambiguity	physical boundary evidence is under-specified
semantic_goal_ambiguity_1	semantic_goal_ambiguity	physical boundary evidence is under-specified
semantic_goal_ambiguity_2	semantic_goal_ambiguity	physical boundary evidence is under-specified
semantic_goal_ambiguity_3	semantic_goal_ambiguity	physical boundary evidence is under-specified
nonlocal_deformable_dynamics_0	nonlocal_deformable_dynamics	needs high-fidelity dynamics
nonlocal_deformable_dynamics_1	nonlocal_deformable_dynamics	needs high-fidelity dynamics
nonlocal_deformable_dynamics_2	nonlocal_deformable_dynamics	needs high-fidelity dynamics
nonlocal_deformable_dynamics_3	nonlocal_deformable_dynamics	needs high-fidelity dynamics
adversarial_helpfulness_0	adversarial_helpfulness	helpfulness must be modeled explicitly
adversarial_helpfulness_1	adversarial_helpfulness	helpfulness must be modeled explicitly
adversarial_helpfulness_2	adversarial_helpfulness	helpfulness must be modeled explicitly
adversarial_helpfulness_3	adversarial_helpfulness	helpfulness must be modeled explicitly
delayed_boundary_shift_0	delayed_boundary_shift	delay can invert the apparent minimal action
delayed_boundary_shift_1	delayed_boundary_shift	delay can invert the apparent minimal action
delayed_boundary_shift_2	delayed_boundary_shift	delay can invert the apparent minimal action
delayed_boundary_shift_3	delayed_boundary_shift	delay can invert the apparent minimal action
under_correction_0	under_correction	pure minimum norm can under-correct
under_correction_1	under_correction	pure minimum norm can under-correct
under_correction_2	under_correction	pure minimum norm can under-correct
under_correction_3	under_correction	pure minimum norm can under-correct

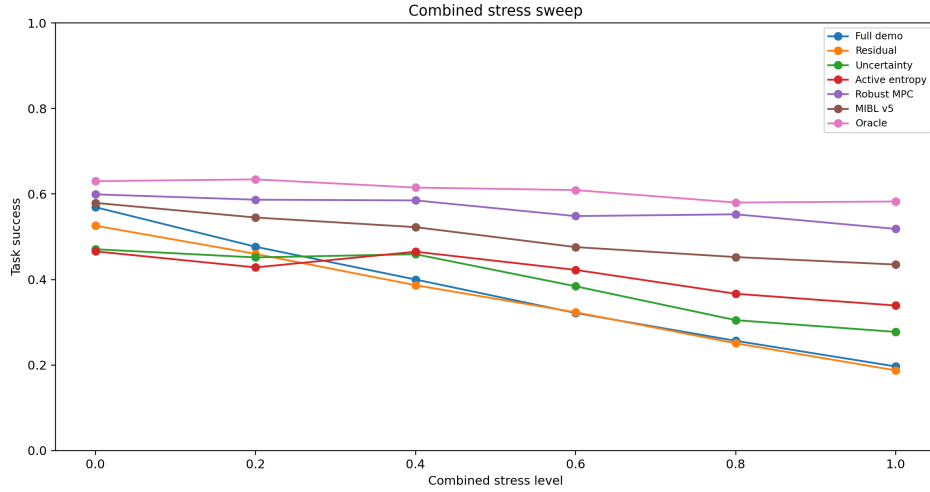


Figure 5: Combined stress sweep. V5 is not dominated at maximum stress, but robust MPC keeps higher success.

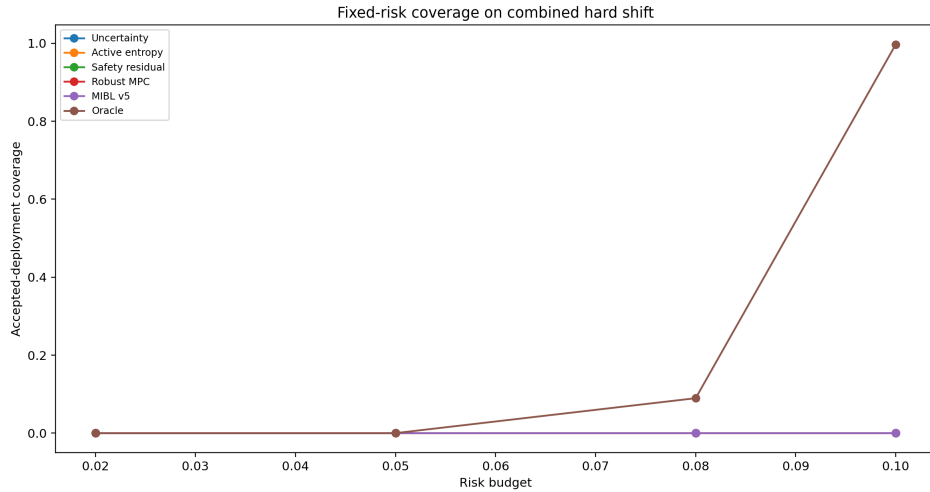


Figure 6: Fixed-risk coverage on combined hard shift. Strict budget 0.05 gives zero coverage.

10 PRIOR-WORK BOUNDARY

The local literature pool is used as a hostile-review pressure map. The paper sits near active correction, residual learning, preference learning, MPC, risk calibration, and robot manipulation datasets (Holzinger et al., 2019; 2017; Wang et al., 2026; Balasubramanian et al., 2012; Yan et al., 2026; Fiorini et al., 2026; Dong et al., 2025). Because this rebuild has no real robot benchmark, it cannot claim to beat those bodies of work directly.

Table 8: Local prior-work pressure map. The entries are used as threat coverage, not as exhaustive manual survey.

Cite	Prior-work pressure	Year	Source
(Yoshida, 2022)	Towards understanding and synthesis of contact-rich anthropomorphic motions through interactive cyber-physical human	2022	Frontiers in Robotics and AI
(Armleder et al., 2022)	Interactive Force Control Based on Multimodal Robot Skin for Physical HumanRobot Collaboration	2022	Advanced Intelligent Systems crossref
(Su, 2026)	Toward Zero-Human-Intervention Autonomous Robot Learning: A Continuous Result-Driven Self-Reward and Correction Framework		
(Zhang et al., 2024)	Human-Robot Interactive Skill Learning and Correction for Polishing Based on Dynamic Time Warping Iterative Learning Con	2024	IEEE Transactions on Control Systems Technology
(Yun & Jun, 2021)	Immersive and interactive cyber-physical system (I2CPS) and virtual reality interface for human involved robotic manufac	2021	Journal of Manufacturing Systems
(Abed et al., 2026)	Formal verification of physical human-robot interaction using interactive theorem proving	2026	Journal of Intelligent Systems
(Holzinger et al., 2019)	A glass-box interactive machine learning approach for solving NP-hard problems with the human-in-the-loop	2019	Creative Mathematics and Informatics
(Holzinger et al., 2017)	A glass-box interactive machine learning approach for solving NP-hard\n problems with the human-in-the-lo	2017	arXiv (Cornell University)
(Wang et al., 2026)	Bidirectional Human-Robot Communication for Physical Human-Robot Interaction	2026	arXiv
(Balasubramanian et al., 2012)	Physical Human Interactive Guidance: Identifying Grasping Principles From Human-Planned Grasps	2012	IEEE Transactions on Robotics
(Yan et al., 2026)	Predicting Tactile Sensory Outcome of Physical Human-Robot Interaction Through Embodied Learning Strategy	2026	IEEE Robotics and Automation Letters
(Fiorini et al., 2026)	Interactive Learning via Physical Human Feedback Using Uncertainty-Aware Energy Tanks	2026	IEEE Robotics and Automation Letters
(Dong et al., 2025)	A 3DGS and LLM-based physical-to-virtual approach for human-robot interactive manufacturing	2025	Manufacturing Letters
(Pool, 2026)	Correction to "ArtistDirectable Motion Generation Using Overlapping Minimum Jerk Trajectories for Interactive Embodied S	2026	Computer Animation and Virtual Worlds
(Wang et al., 2021)	Interactive and Immersive Process-Level Digital Twin for Collaborative Human-Robot Construction Work	2021	Journal of Computing in Civil Engineering
(Wang et al., 2020)	Digital Twin for Human-Robot Interactive Welding and Welder Behavior Analysis	2020	IEEE/CAA Journal of Automatica Sinica
(Song & Xiong, 2025)	Large Language Model-Driven 3D Hyper-Realistic Interactive Intelligent Digital Human System	2025	Sensors
(Katagami & Yamada, 2002a)	Interactive classifier system for real robot learning	2002	openalex
(Holzinger et al., 2018)	Interactive machine learning: experimental evidence for the human in the algorithmic loop	2018	Applied Intelligence
(Huang et al., 2019)	Learning Physical Human-Robot Interaction With Coupled Cooperative Primitives for a Lower Exoskeleton	2019	IEEE Transactions on Automation Science and Engineering
(Mehta & Losey, 2024)	Unified Learning from Demonstrations, Corrections, and Preferences during Physical Human-Robot Interaction	2024	ACM Transactions on Human-Robot Interaction
(Shoukat et al., 2022)	Digital Twin-Driven Virtual Control Technology of Home-Use Robot: Human-Cyber-Physical System	2022	openalex
(Upasani et al., 2023)	Eye-Tracking in Physical Human-Robot Interaction: Mental Workload and Performance Prediction	2023	Human Factors The Journal of the Human Factors and Ergonomic
(Balasubramanian et al., 2014)	Physical Human Interactive Guidance: Identifying Grasping Principles from Human-Planned Grasps	2014	Springer Tracts in Advanced Robotics The Human Hand as an In
(Losey et al., 2022)	Physical interaction as communication: Learning robot objectives online from human corrections	2022	The International Journal of Robotics Research
(Fitter et al., 2020)	Exercising with Baxter: preliminary support for assistive social-physical human-robot interaction	2020	Journal of NeuroEngineering and Rehabilitation
(Haddadin et al., 2008)	Collision Detection and Reaction: A Contribution to Safe Physical Human-Robot Interaction	2008	openalex
(Zhang et al., 2022)	Cooperative Transportation With Mobile Manipulator: A Capability Map-Based Framework for Physical Human-Robot Collaborat	2022	IEEE/ASME Transactions on Mechatronics
(Yin et al., 2020)	Wireless MultistimulusResponsive FabricBased Actuators for Soft Robotic, Human-Machine Interactive, and Wearable Applica	2020	Advanced Materials Technologies

Table 8: Local prior-work pressure map. The entries are used as threat coverage, not as exhaustive manual survey. (continued)

Cite	Prior-work pressure	Year	Source
(Katagami & Yamada, 2002b)	Real Robot Learning with Human Teaching	2002	Tokyo Tech Research Repository (Tokyo Institute of Technology)
(Dincer et al., 2025)	A Machine Learning Approach to Resolving Conflicts in Physical Human-Robot Interaction	2025	ACM Transactions on Human-Robot Interaction
(Regmi et al., 2022a)	Humans modulate arm stiffness to facilitate motor communication during overground physical human-robot interaction	2022	Scientific Reports
(Han & Kim, 2010)	Prediction of the Upper Limb Motion Based on a Geometrical Muscle Changes for Physical Human Machine Interaction	2010	Journal of Institute of Control, Robotics and Systems
(Vianello et al., 2021)	Human Posture Prediction During Physical Human-Robot Interaction	2021	IEEE Robotics and Automation Letters
(Lai et al., 2022)	User intent estimation during robot learning using physical human robot interaction primitives	2022	Autonomous Robots
(You et al., 2024)	Evolution of Industrial Robots from the Perspective of the Metaverse: Integration of Virtual and Physical Realities and Domestic Robots and the Dream of Automation: Understanding Human Interaction and Intervention	2024	Applied Sciences
(Schneiders et al., 2021)	Integrating Perceptions: A Human-Centered Physical Safety Model for Human-Robot Interaction	2021	openalex
(Pandey et al., 2025)	Contact and Physical Interaction	2025	2025 34th IEEE International Conference on Robot and Human I
(Hogan, 2022)	Interactive Learning from Policy-Dependent Human Feedback	2022	Annual Review of Control, Robotics, and Autonomous Systems
(MacGlashan et al., 2017)	Learning from Demonstrations in Human-Robot Collaborative Scenarios: A Survey	2017	arXiv
(Sosa-Ceron et al., 2022)	Deep Reinforcement Learning with Interactive Feedback in a Human-Robot Environment	2022	Robotics
(Moreira et al., 2020b)	Deep Reinforcement Learning with Interactive Feedback in a Human-Robot Environment	2020	Own your potential (DEAKIN)
(Moreira et al., 2020a)	Deep Reinforcement Learning with Interactive Feedback in a Human-Robot Environment	2020	Applied Sciences
(Regmi et al., 2022b)	A robot for overground physical human-robot interaction experiments	2022	PLoS ONE
(Kim et al., 2017)	Intrinsic interactive reinforcement learning - Using error-related potentials for real world human-robot interaction	2017	Scientific Reports
(Lestingi et al., 2023)	Analyzing the impact of human errors on interactive service robotic scenarios via formal verification	2023	Software & Systems Modeling
(d'Elia et al., 2017)	Physical human-robot interaction of an active pelvis orthosis: toward ergonomic assessment of wearable robots	2017	Journal of NeuroEngineering and Rehabilitation
(Pan & Wang, 2026)	Force Sensing Control for Physical Human-Robot Interaction: A Transformer-Based Action Chunking Approach	2026	Machines

11 REPRODUCIBILITY

All tables and figures are regenerated by running `python src/run_experiment.py` from the repository root. The script streams raw CSVs and aggregates seed-level statistics. The canonical local PDF is `C:/Users/wangz/Downloads/85.pdf`; no Desktop copy is part of the artifact contract.

Table 9: Validated row counts for the v5 evidence package.

Artifact	Rows
rollouts.csv	199680
dataset_summary.csv	15360
raw_seed_metrics.csv	1040
metrics.csv	1352
pairwise_stats.csv	768
hard_aggregate_seed_metrics.csv	130
hard_aggregate_metrics.csv	169
hard_aggregate_pairwise_stats.csv	96
ablation_rollouts.csv	33600
ablation_seed_metrics.csv	200
ablation_metrics.csv	20

Table 9: Validated row counts for the v5 evidence package. (continued)

Artifact	Rows
stress_sweep_raw.csv	302400
stress_sweep_seed_metrics.csv	2520
stress_sweep.csv	252
fixed_risk_raw.csv	69120
fixed_risk_seed_metrics.csv	480
fixed_risk_metrics.csv	48
fixed_risk_pairwise.csv	200
negative_cases.csv	24

12 DISCUSSION

The result is not pretty, but it is useful. The v5 learner reduced unsafe overrides relative to most baselines, which supports the idea that safety clipping and calibrated risk matter. But the success and efficiency losses to robust MPC are too large, and the ablations show that the minimum-norm objective is not the causal source of performance in this benchmark. The fastest path to revival is not another synthetic table. The paper would need real human correction traces on robot manipulation tasks, or an accepted high-fidelity benchmark where the correction boundary can be externally verified. It would also need a learned model rather than deterministic proxy components, plus comparisons to modern interactive imitation, preference learning, residual policy correction, and MPC baselines.

13 CONCLUSION

Minimum-intervention human correction remains a compelling research direction. This expanded audit makes the paper more honest and much more rigorous, but it does not make it submission-ready. The correct terminal action is **KILL/ARCHIVE** until external robot or high-fidelity evidence and a mechanism that beats strong baselines are available.

A APPENDIX: FULL MAIN EVIDENCE

This appendix reports every split-method aggregate from the frozen main protocol. The purpose is to make the negative decision auditable: the hard aggregate is not hiding a strong positive result on another predefined split.

Table 10: Full split-by-method main metrics. This appendix table prevents cherry-picking by exposing every predefined split and method.

Split	Method	Success	Efficiency	Damage	Unsafe	Utility
Nominal	No human	0.237 ± 0.011	0.000	0.034	0.030	0.342
Nominal	Full demo	0.565 ± 0.021	1.057	0.043	0.054	0.840
Nominal	DAGger full	0.562 ± 0.023	1.011	0.046	0.060	0.824
Nominal	Residual	0.519 ± 0.017	0.930	0.044	0.055	0.779
Nominal	Preference	0.577 ± 0.026	1.136	0.037	0.051	0.896
Nominal	Uncertainty query	0.493 ± 0.019	0.862	0.033	0.050	0.760
Nominal	Active entropy	0.478 ± 0.032	0.849	0.034	0.036	0.752
Nominal	Safety residual	0.545 ± 0.022	1.043	0.032	0.000	0.855
Nominal	Robust MPC	0.598 ± 0.034	1.268	0.037	0.000	0.950
Nominal	Inverse-RL proxy	0.502 ± 0.019	0.748	0.049	0.067	0.737
Nominal	Min-int v4	0.602 ± 0.020	1.208	0.040	0.057	0.929
Nominal	MIBL v5	0.601 ± 0.028	1.180	0.027	0.000	0.958
Nominal	Oracle minimal	0.626 ± 0.023	1.233	0.029	0.000	0.981
Overcorrect	No human	0.202 ± 0.012	0.000	0.045	0.046	0.296
Overcorrect	Full demo	0.547 ± 0.020	0.914	0.107	0.139	0.718
Overcorrect	DAGger full	0.555 ± 0.014	0.949	0.082	0.101	0.761
Overcorrect	Residual	0.537 ± 0.023	0.949	0.069	0.095	0.766
Overcorrect	Preference	0.554 ± 0.014	1.032	0.034	0.084	0.838
Overcorrect	Uncertainty query	0.448 ± 0.026	0.743	0.053	0.090	0.661
Overcorrect	Active entropy	0.476 ± 0.030	0.810	0.035	0.078	0.724
Overcorrect	Safety residual	0.561 ± 0.024	1.054	0.034	0.001	0.865
Overcorrect	Robust MPC	0.594 ± 0.031	1.176	0.046	0.002	0.915
Overcorrect	Inverse-RL proxy	0.479 ± 0.026	0.702	0.040	0.080	0.701
Overcorrect	Min-int v4	0.576 ± 0.027	1.085	0.055	0.096	0.851
Overcorrect	MIBL v5	0.588 ± 0.031	1.101	0.033	0.000	0.918
Overcorrect	Oracle minimal	0.606 ± 0.017	1.131	0.030	0.000	0.934
Delay	No human	0.158 ± 0.021	0.000	0.037	0.048	0.251
Delay	Full demo	0.469 ± 0.023	0.647	0.092	0.113	0.595
Delay	DAGger full	0.490 ± 0.021	0.682	0.064	0.096	0.646
Delay	Residual	0.441 ± 0.021	0.656	0.053	0.084	0.621
Delay	Preference	0.515 ± 0.020	0.855	0.042	0.082	0.754
Delay	Uncertainty query	0.510 ± 0.023	0.768	0.065	0.111	0.697
Delay	Active entropy	0.457 ± 0.023	0.715	0.045	0.077	0.676
Delay	Safety residual	0.483 ± 0.017	0.777	0.045	0.001	0.722
Delay	Robust MPC	0.608 ± 0.021	1.101	0.041	0.003	0.910
Delay	Inverse-RL proxy	0.465 ± 0.020	0.673	0.046	0.083	0.670
Delay	Min-int v4	0.561 ± 0.029	0.962	0.048	0.099	0.808
Delay	MIBL v5	0.564 ± 0.024	0.963	0.045	0.001	0.843
Delay	Oracle minimal	0.626 ± 0.030	1.063	0.036	0.000	0.929
Ambiguous	No human	0.188 ± 0.015	0.000	0.044	0.044	0.282
Ambiguous	Full demo	0.428 ± 0.023	0.580	0.069	0.092	0.561
Ambiguous	DAGger full	0.414 ± 0.019	0.570	0.057	0.085	0.554
Ambiguous	Residual	0.381 ± 0.014	0.536	0.047	0.083	0.538
Ambiguous	Preference	0.479 ± 0.021	0.808	0.056	0.072	0.696
Ambiguous	Uncertainty query	0.455 ± 0.021	0.678	0.056	0.090	0.633
Ambiguous	Active entropy	0.484 ± 0.027	0.819	0.050	0.067	0.705
Ambiguous	Safety residual	0.437 ± 0.027	0.671	0.038	0.002	0.653
Ambiguous	Robust MPC	0.575 ± 0.022	1.067	0.037	0.002	0.863
Ambiguous	Inverse-RL proxy	0.456 ± 0.006	0.726	0.046	0.069	0.665
Ambiguous	Min-int v4	0.562 ± 0.020	1.007	0.043	0.077	0.817
Ambiguous	MIBL v5	0.561 ± 0.024	0.996	0.037	0.000	0.841
Ambiguous	Oracle minimal	0.585 ± 0.026	1.009	0.037	0.000	0.869
Sparse	No human	0.156 ± 0.015	0.000	0.043	0.052	0.243
Sparse	Full demo	0.377 ± 0.023	0.475	0.055	0.086	0.505
Sparse	DAGger full	0.362 ± 0.020	0.464	0.044	0.076	0.496
Sparse	Residual	0.359 ± 0.016	0.464	0.056	0.069	0.504
Sparse	Preference	0.520 ± 0.029	0.873	0.048	0.079	0.756
Sparse	Uncertainty query	0.398 ± 0.030	0.566	0.041	0.091	0.568
Sparse	Active entropy	0.424 ± 0.015	0.645	0.032	0.064	0.635
Sparse	Safety residual	0.371 ± 0.022	0.542	0.044	0.001	0.561
Sparse	Robust MPC	0.609 ± 0.024	1.113	0.046	0.002	0.906
Sparse	Inverse-RL proxy	0.463 ± 0.026	0.682	0.047	0.075	0.667
Sparse	Min-int v4	0.573 ± 0.025	0.992	0.051	0.083	0.824
Sparse	MIBL v5	0.544 ± 0.024	0.933	0.045	0.000	0.817
Sparse	Oracle minimal	0.600 ± 0.018	1.042	0.040	0.000	0.892
Adversarial	No human	0.151 ± 0.022	0.000	0.043	0.059	0.240
Adversarial	Full demo	0.296 ± 0.009	0.324	0.062	0.099	0.380
Adversarial	DAGger full	0.327 ± 0.025	0.377	0.045	0.108	0.431
Adversarial	Residual	0.311 ± 0.016	0.371	0.046	0.085	0.442
Adversarial	Preference	0.496 ± 0.018	0.782	0.053	0.087	0.709
Adversarial	Uncertainty query	0.360 ± 0.025	0.456	0.057	0.098	0.494
Adversarial	Active entropy	0.416 ± 0.027	0.621	0.046	0.086	0.603
Adversarial	Safety residual	0.319 ± 0.022	0.431	0.034	0.004	0.493
Adversarial	Robust MPC	0.599 ± 0.024	1.024	0.046	0.008	0.876
Adversarial	Inverse-RL proxy	0.452 ± 0.027	0.646	0.046	0.094	0.643
Adversarial	Min-int v4	0.556 ± 0.018	0.917	0.047	0.100	0.789
Adversarial	MIBL v5	0.498 ± 0.021	0.797	0.046	0.002	0.737

Table 10: Full split-by-method main metrics. This appendix table prevents cherry-picking by exposing every predefined split and method. (continued)

Split	Method	Success	Efficiency	Damage	Unsafe	Utility
Adversarial	Oracle minimal	0.604 $\pm$ 0.020	0.982	0.034	0.000	0.886
Dynamics	No human	0.110 $\pm$ 0.012	0.000	0.051	0.094	0.184
Dynamics	Full demo	0.369 $\pm$ 0.022	0.402	0.067	0.143	0.451
Dynamics	DAGger full	0.372 $\pm$ 0.016	0.418	0.051	0.139	0.468
Dynamics	Residual	0.335 $\pm$ 0.024	0.388	0.057	0.135	0.441
Dynamics	Preference	0.452 $\pm$ 0.028	0.595	0.052	0.123	0.611
Dynamics	Uncertainty query	0.413 $\pm$ 0.029	0.494	0.058	0.129	0.538
Dynamics	Active entropy	0.458 $\pm$ 0.015	0.595	0.043	0.109	0.620
Dynamics	Safety residual	0.371 $\pm$ 0.011	0.455	0.041	0.021	0.533
Dynamics	Robust MPC	0.606 $\pm$ 0.028	0.818	0.057	0.023	0.821
Dynamics	Inverse-RL proxy	0.429 $\pm$ 0.019	0.528	0.041	0.137	0.578
Dynamics	Min-int v4	0.550 $\pm$ 0.019	0.748	0.050	0.118	0.731
Dynamics	MIBL v5	0.495 $\pm$ 0.020	0.654	0.051	0.022	0.686
Dynamics	Oracle minimal	0.577 $\pm$ 0.014	0.754	0.041	0.016	0.796
Combined	No human	0.094 $\pm$ 0.013	0.000	0.047	0.076	0.174
Combined	Full demo	0.307 $\pm$ 0.022	0.298	0.073	0.147	0.354
Combined	DAGger full	0.333 $\pm$ 0.026	0.334	0.062	0.142	0.398
Combined	Residual	0.307 $\pm$ 0.021	0.343	0.057	0.137	0.400
Combined	Preference	0.446 $\pm$ 0.028	0.596	0.051	0.135	0.597
Combined	Uncertainty query	0.384 $\pm$ 0.027	0.437	0.059	0.135	0.489
Combined	Active entropy	0.414 $\pm$ 0.012	0.548	0.054	0.109	0.555
Combined	Safety residual	0.332 $\pm$ 0.021	0.407	0.059	0.015	0.473
Combined	Robust MPC	0.565 $\pm$ 0.025	0.800	0.054	0.031	0.772
Combined	Inverse-RL proxy	0.406 $\pm$ 0.016	0.521	0.051	0.129	0.544
Combined	Min-int v4	0.506 $\pm$ 0.014	0.701	0.060	0.153	0.657
Combined	MIBL v5	0.476 $\pm$ 0.022	0.647	0.044	0.014	0.667
Combined	Oracle minimal	0.584 $\pm$ 0.020	0.791	0.047	0.007	0.805

B APPENDIX: HARD-AGGREGATE SEED UNITS

The following seed means are the statistical units used for hard-aggregate paired tests. They show that the negative paired bounds are not a formatting artifact; the baseline advantage appears across seed-level aggregates.

Table 11: Hard-aggregate seed means. These rows are the units used by paired tests.

Method	Seed	Success	Efficiency	Damage	Unsafe	Utility
Active entropy	0	0.427	0.578	0.047	0.097	0.590
Active entropy	1	0.434	0.590	0.042	0.082	0.606
Active entropy	2	0.451	0.630	0.064	0.108	0.613
Active entropy	3	0.431	0.581	0.036	0.106	0.599
Active entropy	4	0.441	0.603	0.056	0.116	0.601
Active entropy	5	0.415	0.577	0.042	0.099	0.579
Active entropy	6	0.448	0.612	0.050	0.111	0.613
Active entropy	7	0.411	0.565	0.045	0.089	0.574
Active entropy	8	0.396	0.544	0.052	0.094	0.550
Active entropy	9	0.438	0.597	0.040	0.115	0.603
DAGger full	0	0.391	0.420	0.043	0.141	0.491
DAGger full	1	0.342	0.383	0.059	0.113	0.434
DAGger full	2	0.372	0.394	0.054	0.125	0.462
DAGger full	3	0.309	0.339	0.061	0.128	0.386
DAGger full	4	0.370	0.400	0.049	0.135	0.464
DAGger full	5	0.349	0.383	0.047	0.127	0.442
DAGger full	6	0.306	0.325	0.054	0.120	0.384
DAGger full	7	0.339	0.379	0.054	0.132	0.425
DAGger full	8	0.340	0.373	0.061	0.135	0.424
DAGger full	9	0.323	0.367	0.045	0.139	0.411
Full demo	0	0.342	0.357	0.066	0.132	0.417
Full demo	1	0.311	0.320	0.069	0.130	0.376
Full demo	2	0.318	0.353	0.073	0.128	0.387
Full demo	3	0.328	0.324	0.052	0.137	0.402
Full demo	4	0.335	0.364	0.059	0.127	0.417
Full demo	5	0.311	0.334	0.066	0.109	0.386
Full demo	6	0.302	0.318	0.080	0.141	0.358
Full demo	7	0.345	0.371	0.087	0.134	0.410
Full demo	8	0.302	0.314	0.066	0.120	0.373
Full demo	9	0.345	0.360	0.056	0.141	0.424
Inverse-RL proxy	0	0.415	0.540	0.047	0.113	0.570
Inverse-RL proxy	1	0.422	0.537	0.052	0.123	0.570
Inverse-RL proxy	2	0.446	0.604	0.059	0.134	0.603
Inverse-RL proxy	3	0.434	0.568	0.045	0.115	0.596
Inverse-RL proxy	4	0.410	0.549	0.054	0.104	0.567
Inverse-RL proxy	5	0.439	0.580	0.047	0.137	0.596
Inverse-RL proxy	6	0.443	0.587	0.042	0.095	0.616
Inverse-RL proxy	7	0.425	0.549	0.023	0.137	0.588
Inverse-RL proxy	8	0.408	0.549	0.038	0.113	0.570

Table 11: Hard-aggregate seed means. These rows are the units used by paired tests. (continued)

Method	Seed	Success	Efficiency	Damage	Unsafe	Utility
Inverse-RL proxy	9	0.446	0.587	0.052	0.130	0.604
MIBL v5	0	0.491	0.684	0.050	0.007	0.696
MIBL v5	1	0.476	0.671	0.049	0.014	0.674
MIBL v5	2	0.526	0.769	0.052	0.009	0.747
MIBL v5	3	0.497	0.705	0.045	0.019	0.705
MIBL v5	4	0.490	0.723	0.045	0.014	0.703
MIBL v5	5	0.460	0.652	0.059	0.005	0.652
MIBL v5	6	0.479	0.684	0.031	0.007	0.693
MIBL v5	7	0.497	0.704	0.035	0.021	0.708
MIBL v5	8	0.493	0.704	0.050	0.012	0.700
MIBL v5	9	0.488	0.695	0.052	0.017	0.690
Min-int v4	0	0.530	0.772	0.076	0.122	0.702
Min-int v4	1	0.536	0.773	0.068	0.106	0.717
Min-int v4	2	0.509	0.754	0.047	0.120	0.692
Min-int v4	3	0.533	0.781	0.043	0.109	0.729
Min-int v4	4	0.545	0.806	0.049	0.127	0.740
Min-int v4	5	0.554	0.807	0.047	0.130	0.746
Min-int v4	6	0.536	0.799	0.059	0.108	0.728
Min-int v4	7	0.552	0.808	0.054	0.142	0.738
Min-int v4	8	0.556	0.808	0.045	0.155	0.743
Min-int v4	9	0.523	0.780	0.036	0.118	0.719
No human	0	0.111	0.000	0.045	0.076	0.193
No human	1	0.090	0.000	0.035	0.073	0.178
No human	2	0.141	0.000	0.056	0.092	0.213
No human	3	0.104	0.000	0.033	0.078	0.192
No human	4	0.116	0.000	0.045	0.064	0.201
No human	5	0.151	0.000	0.042	0.101	0.228
No human	6	0.137	0.000	0.062	0.061	0.214
No human	7	0.109	0.000	0.043	0.080	0.191
No human	8	0.116	0.000	0.054	0.061	0.198
No human	9	0.108	0.000	0.054	0.078	0.184
Oracle minimal	0	0.611	0.867	0.031	0.007	0.864
Oracle minimal	1	0.613	0.863	0.052	0.009	0.851
Oracle minimal	2	0.580	0.847	0.033	0.010	0.825
Oracle minimal	3	0.611	0.871	0.036	0.003	0.862
Oracle minimal	4	0.559	0.816	0.040	0.009	0.795
Oracle minimal	5	0.601	0.855	0.038	0.010	0.844
Oracle minimal	6	0.554	0.795	0.049	0.005	0.781
Oracle minimal	7	0.590	0.842	0.047	0.010	0.826
Oracle minimal	8	0.575	0.821	0.038	0.005	0.813
Oracle minimal	9	0.589	0.843	0.043	0.007	0.828
Preference	0	0.493	0.690	0.056	0.104	0.675
Preference	1	0.411	0.570	0.062	0.099	0.565
Preference	2	0.465	0.672	0.049	0.104	0.647
Preference	3	0.477	0.678	0.040	0.109	0.664
Preference	4	0.470	0.643	0.056	0.115	0.640
Preference	5	0.443	0.630	0.054	0.130	0.605
Preference	6	0.460	0.671	0.038	0.113	0.644
Preference	7	0.490	0.697	0.056	0.111	0.671
Preference	8	0.448	0.641	0.056	0.127	0.613
Preference	9	0.491	0.685	0.052	0.139	0.665
Residual	0	0.323	0.380	0.056	0.111	0.438
Residual	1	0.311	0.358	0.056	0.111	0.421
Residual	2	0.321	0.373	0.049	0.104	0.437
Residual	3	0.290	0.339	0.056	0.128	0.391
Residual	4	0.309	0.365	0.047	0.109	0.425
Residual	5	0.326	0.369	0.050	0.122	0.436
Residual	6	0.295	0.327	0.061	0.127	0.390
Residual	7	0.344	0.398	0.061	0.134	0.451
Residual	8	0.347	0.402	0.040	0.137	0.469
Residual	9	0.311	0.361	0.061	0.108	0.418
Robust MPC	0	0.620	0.907	0.057	0.014	0.859
Robust MPC	1	0.576	0.846	0.050	0.024	0.800
Robust MPC	2	0.585	0.884	0.061	0.016	0.816
Robust MPC	3	0.594	0.880	0.047	0.017	0.831
Robust MPC	4	0.611	0.928	0.031	0.021	0.866
Robust MPC	5	0.583	0.865	0.069	0.026	0.802
Robust MPC	6	0.594	0.898	0.043	0.026	0.833
Robust MPC	7	0.606	0.917	0.052	0.021	0.846
Robust MPC	8	0.536	0.793	0.057	0.021	0.747
Robust MPC	9	0.595	0.886	0.056	0.021	0.827
Safety residual	0	0.352	0.441	0.031	0.019	0.519
Safety residual	1	0.351	0.427	0.052	0.014	0.504
Safety residual	2	0.339	0.442	0.047	0.014	0.497
Safety residual	3	0.342	0.433	0.043	0.012	0.503
Safety residual	4	0.358	0.456	0.033	0.024	0.526
Safety residual	5	0.340	0.438	0.061	0.010	0.492
Safety residual	6	0.304	0.385	0.049	0.009	0.452
Safety residual	7	0.345	0.438	0.030	0.010	0.513
Safety residual	8	0.311	0.384	0.050	0.012	0.458
Safety residual	9	0.368	0.466	0.049	0.009	0.533
Uncertainty query	0	0.385	0.464	0.054	0.120	0.509
Uncertainty query	1	0.382	0.443	0.038	0.115	0.511
Uncertainty query	2	0.410	0.487	0.038	0.132	0.542
Uncertainty query	3	0.377	0.444	0.069	0.118	0.489
Uncertainty query	4	0.370	0.449	0.052	0.118	0.493
Uncertainty query	5	0.384	0.471	0.071	0.118	0.501
Uncertainty query	6	0.378	0.447	0.061	0.111	0.497
Uncertainty query	7	0.413	0.502	0.076	0.141	0.527

Table 11: Hard-aggregate seed means. These rows are the units used by paired tests. (continued)

Method	Seed	Success	Efficiency	Damage	Unsafe	Utility
Uncertainty query	8	0.333	0.404	0.057	0.122	0.444
Uncertainty query	9	0.425	0.511	0.062	0.113	0.557

C APPENDIX: COMPLETE PAIRED TESTS

The table below gives every hard-aggregate paired difference for v5 minus each reference method across the predefined paired metrics. Primary success and efficiency comparisons fail against robust MPC and v4.

Table 12: All hard-aggregate paired differences for MIBL v5 minus each reference method.

Reference	Metric	Mean diff	CI95	Lower95
Active entropy	task_success	0.060	0.013	0.048
Active entropy	correction_efficiency	0.111	0.019	0.093
Active entropy	damage	-0.001	0.007	-0.008
Active entropy	intervention_magnitude	0.065	0.002	0.063
Active entropy	boundary_error	-0.104	0.004	-0.108
Active entropy	intent_preservation	0.006	0.001	0.005
Active entropy	unsafe_override	-0.089	0.008	-0.097
Active entropy	robust_utility	0.104	0.017	0.087
DAgger full	task_success	0.146	0.018	0.128
DAgger full	correction_efficiency	0.323	0.024	0.299
DAgger full	damage	-0.006	0.007	-0.013
DAgger full	intervention_magnitude	0.032	0.005	0.026
DAgger full	boundary_error	-0.300	0.010	-0.309
DAgger full	intent_preservation	-0.015	0.003	-0.018
DAgger full	unsafe_override	-0.117	0.006	-0.123
DAgger full	robust_utility	0.264	0.024	0.240
Full demo	task_success	0.166	0.013	0.153
Full demo	correction_efficiency	0.358	0.019	0.338
Full demo	damage	-0.020	0.010	-0.031
Full demo	intervention_magnitude	-0.001	0.006	-0.007
Full demo	boundary_error	-0.335	0.010	-0.344
Full demo	intent_preservation	-0.021	0.003	-0.024
Full demo	unsafe_override	-0.117	0.005	-0.123
Full demo	robust_utility	0.302	0.019	0.283
Inverse-RL proxy	task_success	0.061	0.014	0.047
Inverse-RL proxy	correction_efficiency	0.134	0.020	0.114
Inverse-RL proxy	damage	0.001	0.005	-0.004
Inverse-RL proxy	intervention_magnitude	0.052	0.002	0.050
Inverse-RL proxy	boundary_error	-0.101	0.005	-0.105
Inverse-RL proxy	intent_preservation	-0.055	0.002	-0.057
Inverse-RL proxy	unsafe_override	-0.108	0.009	-0.117
Inverse-RL proxy	robust_utility	0.109	0.018	0.091
Min-int v4	task_success	-0.048	0.018	-0.065
Min-int v4	correction_efficiency	-0.090	0.027	-0.116
Min-int v4	damage	-0.006	0.010	-0.016
Min-int v4	intervention_magnitude	-0.053	0.002	-0.054
Min-int v4	boundary_error	0.013	0.006	0.007
Min-int v4	intent_preservation	0.004	0.002	0.003
Min-int v4	unsafe_override	-0.111	0.010	-0.121
Min-int v4	robust_utility	-0.029	0.023	-0.052
No human	task_success	0.371	0.016	0.355
No human	correction_efficiency	0.699	0.020	0.680
No human	damage	0.000	0.009	-0.009
No human	intervention_magnitude	0.460	0.004	0.456
No human	boundary_error	-0.809	0.003	-0.812
No human	intent_preservation	-0.035	0.003	-0.038
No human	unsafe_override	-0.064	0.009	-0.073
No human	robust_utility	0.497	0.018	0.479
Oracle minimal	task_success	-0.099	0.018	-0.117
Oracle minimal	correction_efficiency	-0.143	0.026	-0.170
Oracle minimal	damage	0.006	0.008	-0.002
Oracle minimal	intervention_magnitude	-0.029	0.002	-0.031
Oracle minimal	boundary_error	0.064	0.003	0.060
Oracle minimal	intent_preservation	0.000	0.000	0.000
Oracle minimal	unsafe_override	0.005	0.004	0.001
Oracle minimal	robust_utility	-0.132	0.025	-0.157
Preference	task_success	0.025	0.015	0.010
Preference	correction_efficiency	0.041	0.025	0.017
Preference	damage	-0.005	0.005	-0.010
Preference	intervention_magnitude	0.043	0.001	0.042
Preference	boundary_error	-0.047	0.004	-0.050
Preference	intent_preservation	0.021	0.001	0.020
Preference	unsafe_override	-0.103	0.009	-0.111
Preference	robust_utility	0.058	0.019	0.038
Residual	task_success	0.172	0.015	0.157

Table 12: All hard-aggregate paired differences for MIBL v5 minus each reference method. (continued)

Reference	Metric	Mean diff	CI95	Lower95
Residual	correction_efficiency	0.332	0.022	0.310
Residual	damage	-0.007	0.008	-0.015
Residual	intervention_magnitude	0.125	0.006	0.119
Residual	boundary_error	-0.343	0.011	-0.353
Residual	intent_preservation	0.005	0.004	0.001
Residual	unsafe_override	-0.107	0.007	-0.114
Residual	robust_utility	0.269	0.020	0.249
Robust MPC	task_success	-0.101	0.017	-0.118
Robust MPC	correction_efficiency	-0.181	0.028	-0.209
Robust MPC	damage	-0.006	0.005	-0.011
Robust MPC	intervention_magnitude	-0.077	0.002	-0.079
Robust MPC	boundary_error	0.081	0.003	0.078
Robust MPC	intent_preservation	0.021	0.000	0.021
Robust MPC	unsafe_override	-0.008	0.005	-0.013
Robust MPC	robust_utility	-0.126	0.024	-0.150
Safety residual	task_success	0.149	0.016	0.133
Safety residual	correction_efficiency	0.268	0.023	0.245
Safety residual	damage	0.002	0.006	-0.004
Safety residual	intervention_magnitude	0.152	0.004	0.148
Safety residual	boundary_error	-0.246	0.007	-0.253
Safety residual	intent_preservation	0.026	0.001	0.025
Safety residual	unsafe_override	-0.001	0.005	-0.006
Safety residual	robust_utility	0.197	0.022	0.175
Uncertainty query	task_success	0.104	0.017	0.087
Uncertainty query	correction_efficiency	0.237	0.026	0.211
Uncertainty query	damage	-0.011	0.011	-0.022
Uncertainty query	intervention_magnitude	0.031	0.004	0.027
Uncertainty query	boundary_error	-0.187	0.007	-0.194
Uncertainty query	intent_preservation	-0.002	0.003	-0.005
Uncertainty query	unsafe_override	-0.108	0.006	-0.114
Uncertainty query	robust_utility	0.190	0.022	0.168

D APPENDIX: COMPLETE STRESS SWEEP

The main text highlights combined stress because it is the most important deployment stressor, but all axes are reported here. This table includes human noise, correction delay, overcorrection bias, intent ambiguity, helpfulness shift, and combined stress.

Table 13: All stress-sweep metrics across six axes, six levels, and seven methods.

Axis	Level	Method	Success	Eff.	Damage	Unsafe	Utility
human_noise	0.0	Full demo	0.336	0.393	0.049	0.136	0.430
human_noise	0.0	Residual	0.309	0.360	0.047	0.138	0.413
human_noise	0.0	Uncertainty query	0.386	0.486	0.053	0.125	0.512
human_noise	0.0	Active entropy	0.411	0.551	0.054	0.112	0.553
human_noise	0.0	Robust MPC	0.566	0.818	0.051	0.029	0.778
human_noise	0.0	MIBL v5	0.487	0.676	0.046	0.012	0.684
human_noise	0.0	Oracle minimal	0.578	0.795	0.048	0.006	0.801
human_noise	0.2	Full demo	0.321	0.356	0.058	0.117	0.405
human_noise	0.2	Residual	0.314	0.359	0.048	0.138	0.418
human_noise	0.2	Uncertainty query	0.380	0.465	0.054	0.134	0.498
human_noise	0.2	Active entropy	0.415	0.551	0.055	0.103	0.559
human_noise	0.2	Robust MPC	0.571	0.812	0.050	0.027	0.783
human_noise	0.2	MIBL v5	0.485	0.669	0.048	0.016	0.678
human_noise	0.2	Oracle minimal	0.590	0.811	0.048	0.004	0.816
human_noise	0.4	Full demo	0.302	0.305	0.068	0.142	0.356
human_noise	0.4	Residual	0.294	0.330	0.046	0.143	0.389
human_noise	0.4	Uncertainty query	0.373	0.423	0.058	0.131	0.478
human_noise	0.4	Active entropy	0.394	0.527	0.055	0.102	0.533
human_noise	0.4	Robust MPC	0.567	0.794	0.053	0.028	0.773
human_noise	0.4	MIBL v5	0.480	0.657	0.052	0.018	0.668
human_noise	0.4	Oracle minimal	0.592	0.805	0.047	0.007	0.816
human_noise	0.6	Full demo	0.297	0.277	0.082	0.155	0.327
human_noise	0.6	Residual	0.285	0.316	0.053	0.149	0.371
human_noise	0.6	Uncertainty query	0.367	0.394	0.062	0.146	0.455
human_noise	0.6	Active entropy	0.403	0.530	0.053	0.105	0.542
human_noise	0.6	Robust MPC	0.556	0.778	0.051	0.032	0.759
human_noise	0.6	MIBL v5	0.472	0.628	0.052	0.016	0.654
human_noise	0.6	Oracle minimal	0.580	0.780	0.048	0.009	0.797
human_noise	0.8	Full demo	0.318	0.272	0.122	0.178	0.311
human_noise	0.8	Residual	0.297	0.337	0.060	0.135	0.386
human_noise	0.8	Uncertainty query	0.347	0.365	0.085	0.160	0.407
human_noise	0.8	Active entropy	0.391	0.530	0.058	0.112	0.526
human_noise	0.8	Robust MPC	0.543	0.779	0.050	0.030	0.749
human_noise	0.8	MIBL v5	0.455	0.627	0.050	0.017	0.638
human_noise	0.8	Oracle minimal	0.588	0.806	0.046	0.006	0.813
human_noise	1.0	Full demo	0.276	0.225	0.150	0.204	0.232

Table 13: All stress-sweep metrics across six axes, six levels, and seven methods. (continued)

Axis	Level	Method	Success	Eff.	Damage	Unsafe	Utility
human_noise	1.0	Residual	0.284	0.317	0.060	0.147	0.365
human_noise	1.0	Uncertainty query	0.349	0.340	0.094	0.158	0.397
human_noise	1.0	Active entropy	0.408	0.533	0.053	0.105	0.550
human_noise	1.0	Robust MPC	0.538	0.750	0.052	0.028	0.738
human_noise	1.0	MIBL v5	0.466	0.629	0.047	0.017	0.652
human_noise	1.0	Oracle minimal	0.595	0.818	0.052	0.007	0.820
correction_delay	0.0	Full demo	0.331	0.330	0.062	0.131	0.401
correction_delay	0.0	Residual	0.297	0.343	0.045	0.142	0.401
correction_delay	0.0	Uncertainty query	0.364	0.426	0.054	0.131	0.476
correction_delay	0.0	Active entropy	0.405	0.533	0.056	0.103	0.548
correction_delay	0.0	Robust MPC	0.568	0.799	0.050	0.028	0.781
correction_delay	0.0	MIBL v5	0.468	0.639	0.049	0.013	0.658
correction_delay	0.0	Oracle minimal	0.588	0.806	0.046	0.007	0.816
correction_delay	0.2	Full demo	0.326	0.334	0.062	0.143	0.393
correction_delay	0.2	Residual	0.312	0.346	0.048	0.134	0.416
correction_delay	0.2	Uncertainty query	0.373	0.429	0.054	0.136	0.483
correction_delay	0.2	Active entropy	0.402	0.527	0.053	0.104	0.543
correction_delay	0.2	Robust MPC	0.566	0.791	0.048	0.028	0.776
correction_delay	0.2	MIBL v5	0.477	0.642	0.048	0.009	0.668
correction_delay	0.2	Oracle minimal	0.589	0.802	0.045	0.007	0.816
correction_delay	0.4	Full demo	0.307	0.316	0.068	0.141	0.365
correction_delay	0.4	Residual	0.289	0.327	0.049	0.137	0.385
correction_delay	0.4	Uncertainty query	0.361	0.416	0.057	0.138	0.463
correction_delay	0.4	Active entropy	0.396	0.527	0.055	0.110	0.533
correction_delay	0.4	Robust MPC	0.576	0.812	0.048	0.029	0.789
correction_delay	0.4	MIBL v5	0.483	0.651	0.048	0.015	0.674
correction_delay	0.4	Oracle minimal	0.583	0.793	0.048	0.007	0.804
correction_delay	0.6	Full demo	0.338	0.345	0.067	0.141	0.400
correction_delay	0.6	Residual	0.306	0.342	0.047	0.143	0.403
correction_delay	0.6	Uncertainty query	0.364	0.417	0.063	0.138	0.462
correction_delay	0.6	Active entropy	0.398	0.530	0.052	0.109	0.537
correction_delay	0.6	Robust MPC	0.570	0.819	0.052	0.032	0.781
correction_delay	0.6	MIBL v5	0.473	0.652	0.049	0.016	0.662
correction_delay	0.6	Oracle minimal	0.598	0.820	0.046	0.002	0.828
correction_delay	0.8	Full demo	0.300	0.295	0.082	0.157	0.336
correction_delay	0.8	Residual	0.292	0.334	0.053	0.144	0.382
correction_delay	0.8	Uncertainty query	0.362	0.408	0.060	0.142	0.457
correction_delay	0.8	Active entropy	0.407	0.539	0.055	0.102	0.547
correction_delay	0.8	Robust MPC	0.567	0.804	0.049	0.028	0.776
correction_delay	0.8	MIBL v5	0.477	0.651	0.050	0.014	0.663
correction_delay	0.8	Oracle minimal	0.593	0.812	0.050	0.007	0.816
correction_delay	1.0	Full demo	0.298	0.284	0.085	0.147	0.328
correction_delay	1.0	Residual	0.288	0.317	0.054	0.138	0.372
correction_delay	1.0	Uncertainty query	0.372	0.408	0.065	0.142	0.459
correction_delay	1.0	Active entropy	0.417	0.559	0.058	0.103	0.558
correction_delay	1.0	Robust MPC	0.551	0.774	0.053	0.026	0.750
correction_delay	1.0	MIBL v5	0.470	0.649	0.052	0.014	0.653
correction_delay	1.0	Oracle minimal	0.575	0.789	0.050	0.010	0.789
overcorrection_bias	0.0	Full demo	0.247	0.224	0.071	0.120	0.293
overcorrection_bias	0.0	Residual	0.235	0.244	0.048	0.122	0.321
overcorrection_bias	0.0	Uncertainty query	0.323	0.349	0.058	0.120	0.418
overcorrection_bias	0.0	Active entropy	0.357	0.464	0.054	0.098	0.488
overcorrection_bias	0.0	Robust MPC	0.563	0.814	0.049	0.022	0.778
overcorrection_bias	0.0	MIBL v5	0.473	0.653	0.048	0.007	0.667
overcorrection_bias	0.0	Oracle minimal	0.597	0.829	0.048	0.005	0.826
overcorrection_bias	0.2	Full demo	0.288	0.268	0.063	0.128	0.343
overcorrection_bias	0.2	Residual	0.245	0.265	0.042	0.128	0.336
overcorrection_bias	0.2	Uncertainty query	0.330	0.365	0.052	0.127	0.428
overcorrection_bias	0.2	Active entropy	0.366	0.486	0.056	0.107	0.497
overcorrection_bias	0.2	Robust MPC	0.564	0.794	0.052	0.028	0.771
overcorrection_bias	0.2	MIBL v5	0.459	0.627	0.048	0.013	0.645
overcorrection_bias	0.2	Oracle minimal	0.583	0.789	0.049	0.006	0.803
overcorrection_bias	0.4	Full demo	0.297	0.279	0.068	0.146	0.345
overcorrection_bias	0.4	Residual	0.269	0.309	0.049	0.131	0.363
overcorrection_bias	0.4	Uncertainty query	0.365	0.418	0.053	0.119	0.475
overcorrection_bias	0.4	Active entropy	0.388	0.516	0.057	0.103	0.525
overcorrection_bias	0.4	Robust MPC	0.576	0.820	0.052	0.029	0.789
overcorrection_bias	0.4	MIBL v5	0.485	0.666	0.048	0.013	0.679
overcorrection_bias	0.4	Oracle minimal	0.617	0.848	0.047	0.004	0.852
overcorrection_bias	0.6	Full demo	0.323	0.324	0.067	0.147	0.380
overcorrection_bias	0.6	Residual	0.283	0.326	0.050	0.150	0.375
overcorrection_bias	0.6	Uncertainty query	0.389	0.448	0.060	0.143	0.495
overcorrection_bias	0.6	Active entropy	0.418	0.556	0.059	0.107	0.560
overcorrection_bias	0.6	Robust MPC	0.570	0.811	0.054	0.032	0.779
overcorrection_bias	0.6	MIBL v5	0.480	0.653	0.049	0.014	0.669
overcorrection_bias	0.6	Oracle minimal	0.588	0.802	0.048	0.008	0.810
overcorrection_bias	0.8	Full demo	0.343	0.353	0.069	0.147	0.403
overcorrection_bias	0.8	Residual	0.328	0.372	0.047	0.152	0.426
overcorrection_bias	0.8	Uncertainty query	0.399	0.466	0.065	0.145	0.503
overcorrection_bias	0.8	Active entropy	0.432	0.576	0.055	0.118	0.574
overcorrection_bias	0.8	Robust MPC	0.568	0.794	0.053	0.033	0.771
overcorrection_bias	0.8	MIBL v5	0.496	0.663	0.052	0.015	0.685
overcorrection_bias	0.8	Oracle minimal	0.598	0.798	0.049	0.003	0.821
overcorrection_bias	1.0	Full demo	0.352	0.360	0.085	0.163	0.397
overcorrection_bias	1.0	Residual	0.333	0.394	0.049	0.150	0.434
overcorrection_bias	1.0	Uncertainty query	0.404	0.473	0.062	0.157	0.507
overcorrection_bias	1.0	Active entropy	0.439	0.597	0.053	0.108	0.590
overcorrection_bias	1.0	Robust MPC	0.564	0.805	0.052	0.027	0.774
overcorrection_bias	1.0	MIBL v5	0.479	0.666	0.050	0.019	0.669

Table 13: All stress-sweep metrics across six axes, six levels, and seven methods. (continued)

Axis	Level	Method	Success	Eff.	Damage	Unsafe	Utility
overcorrection_bias	1.0	Oracle minimal	0.589	0.805	0.049	0.009	0.812
intent_ambiguity	0.0	Full demo	0.374	0.396	0.071	0.139	0.451
intent_ambiguity	0.0	Residual	0.354	0.422	0.045	0.142	0.475
intent_ambiguity	0.0	Uncertainty query	0.429	0.507	0.058	0.143	0.554
intent_ambiguity	0.0	Active entropy	0.349	0.447	0.052	0.105	0.497
intent_ambiguity	0.0	Robust MPC	0.568	0.797	0.047	0.032	0.788
intent_ambiguity	0.0	MIBL v5	0.487	0.657	0.045	0.018	0.694
intent_ambiguity	0.0	Oracle minimal	0.601	0.809	0.042	0.007	0.839
intent_ambiguity	0.2	Full demo	0.341	0.358	0.075	0.149	0.402
intent_ambiguity	0.2	Residual	0.343	0.413	0.049	0.137	0.459
intent_ambiguity	0.2	Uncertainty query	0.406	0.477	0.061	0.143	0.521
intent_ambiguity	0.2	Active entropy	0.432	0.575	0.049	0.104	0.591
intent_ambiguity	0.2	Robust MPC	0.569	0.800	0.046	0.025	0.789
intent_ambiguity	0.2	MIBL v5	0.495	0.670	0.045	0.014	0.699
intent_ambiguity	0.2	Oracle minimal	0.603	0.812	0.043	0.004	0.839
intent_ambiguity	0.4	Full demo	0.333	0.340	0.072	0.138	0.395
intent_ambiguity	0.4	Residual	0.297	0.342	0.049	0.148	0.394
intent_ambiguity	0.4	Uncertainty query	0.383	0.440	0.062	0.131	0.491
intent_ambiguity	0.4	Active entropy	0.410	0.540	0.053	0.105	0.555
intent_ambiguity	0.4	Robust MPC	0.571	0.799	0.049	0.030	0.783
intent_ambiguity	0.4	MIBL v5	0.479	0.649	0.048	0.019	0.671
intent_ambiguity	0.4	Oracle minimal	0.593	0.802	0.048	0.008	0.818
intent_ambiguity	0.6	Full demo	0.302	0.300	0.072	0.136	0.354
intent_ambiguity	0.6	Residual	0.291	0.327	0.049	0.141	0.384
intent_ambiguity	0.6	Uncertainty query	0.363	0.410	0.058	0.130	0.465
intent_ambiguity	0.6	Active entropy	0.408	0.537	0.055	0.108	0.548
intent_ambiguity	0.6	Robust MPC	0.562	0.791	0.054	0.030	0.766
intent_ambiguity	0.6	MIBL v5	0.468	0.633	0.049	0.014	0.652
intent_ambiguity	0.6	Oracle minimal	0.592	0.812	0.048	0.005	0.817
intent_ambiguity	0.8	Full demo	0.281	0.271	0.068	0.146	0.324
intent_ambiguity	0.8	Residual	0.267	0.297	0.048	0.142	0.352
intent_ambiguity	0.8	Uncertainty query	0.352	0.395	0.059	0.138	0.446
intent_ambiguity	0.8	Active entropy	0.388	0.521	0.054	0.103	0.522
intent_ambiguity	0.8	Robust MPC	0.566	0.812	0.054	0.033	0.770
intent_ambiguity	0.8	MIBL v5	0.476	0.656	0.049	0.017	0.661
intent_ambiguity	0.8	Oracle minimal	0.593	0.821	0.050	0.006	0.817
intent_ambiguity	1.0	Full demo	0.268	0.255	0.071	0.132	0.309
intent_ambiguity	1.0	Residual	0.263	0.289	0.050	0.136	0.347
intent_ambiguity	1.0	Uncertainty query	0.321	0.350	0.062	0.136	0.401
intent_ambiguity	1.0	Active entropy	0.375	0.498	0.059	0.110	0.496
intent_ambiguity	1.0	Robust MPC	0.560	0.800	0.056	0.033	0.757
intent_ambiguity	1.0	MIBL v5	0.460	0.639	0.052	0.016	0.636
intent_ambiguity	1.0	Oracle minimal	0.576	0.794	0.052	0.005	0.788
helpfulness_shift	0.0	Full demo	0.408	0.424	0.101	0.159	0.452
helpfulness_shift	0.0	Residual	0.389	0.467	0.067	0.144	0.490
helpfulness_shift	0.0	Uncertainty query	0.458	0.532	0.077	0.148	0.560
helpfulness_shift	0.0	Active entropy	0.451	0.612	0.062	0.111	0.594
helpfulness_shift	0.0	Robust MPC	0.569	0.801	0.057	0.028	0.774
helpfulness_shift	0.0	MIBL v5	0.511	0.699	0.052	0.016	0.703
helpfulness_shift	0.0	Oracle minimal	0.590	0.805	0.048	0.006	0.813
helpfulness_shift	0.2	Full demo	0.388	0.411	0.085	0.153	0.444
helpfulness_shift	0.2	Residual	0.381	0.459	0.055	0.154	0.489
helpfulness_shift	0.2	Uncertainty query	0.432	0.519	0.073	0.142	0.541
helpfulness_shift	0.2	Active entropy	0.458	0.637	0.058	0.107	0.613
helpfulness_shift	0.2	Robust MPC	0.574	0.814	0.048	0.026	0.788
helpfulness_shift	0.2	MIBL v5	0.514	0.714	0.052	0.015	0.712
helpfulness_shift	0.2	Oracle minimal	0.587	0.791	0.045	0.006	0.809
helpfulness_shift	0.4	Full demo	0.352	0.361	0.069	0.151	0.411
helpfulness_shift	0.4	Residual	0.333	0.387	0.054	0.139	0.436
helpfulness_shift	0.4	Uncertainty query	0.403	0.465	0.063	0.138	0.510
helpfulness_shift	0.4	Active entropy	0.424	0.576	0.056	0.106	0.570
helpfulness_shift	0.4	Robust MPC	0.564	0.788	0.048	0.035	0.770
helpfulness_shift	0.4	MIBL v5	0.487	0.663	0.049	0.016	0.677
helpfulness_shift	0.4	Oracle minimal	0.593	0.800	0.046	0.007	0.817
helpfulness_shift	0.6	Full demo	0.318	0.322	0.070	0.140	0.375
helpfulness_shift	0.6	Residual	0.295	0.329	0.048	0.140	0.390
helpfulness_shift	0.6	Uncertainty query	0.364	0.416	0.062	0.146	0.461
helpfulness_shift	0.6	Active entropy	0.395	0.526	0.056	0.111	0.531
helpfulness_shift	0.6	Robust MPC	0.564	0.798	0.048	0.028	0.774
helpfulness_shift	0.6	MIBL v5	0.481	0.650	0.047	0.018	0.670
helpfulness_shift	0.6	Oracle minimal	0.596	0.821	0.048	0.005	0.824
helpfulness_shift	0.8	Full demo	0.253	0.229	0.062	0.147	0.295
helpfulness_shift	0.8	Residual	0.240	0.257	0.047	0.136	0.325
helpfulness_shift	0.8	Uncertainty query	0.328	0.361	0.056	0.149	0.417
helpfulness_shift	0.8	Active entropy	0.378	0.497	0.053	0.105	0.511
helpfulness_shift	0.8	Robust MPC	0.568	0.801	0.052	0.028	0.775
helpfulness_shift	0.8	MIBL v5	0.458	0.630	0.051	0.013	0.642
helpfulness_shift	0.8	Oracle minimal	0.580	0.792	0.052	0.008	0.798
helpfulness_shift	1.0	Full demo	0.172	0.130	0.055	0.142	0.200
helpfulness_shift	1.0	Residual	0.181	0.158	0.046	0.145	0.246
helpfulness_shift	1.0	Uncertainty query	0.271	0.265	0.056	0.144	0.343
helpfulness_shift	1.0	Active entropy	0.333	0.420	0.057	0.107	0.451
helpfulness_shift	1.0	Robust MPC	0.564	0.793	0.052	0.027	0.769
helpfulness_shift	1.0	MIBL v5	0.429	0.577	0.047	0.013	0.604
helpfulness_shift	1.0	Oracle minimal	0.587	0.799	0.048	0.007	0.807
combined	0.0	Full demo	0.569	1.064	0.040	0.048	0.852
combined	0.0	Residual	0.526	0.923	0.039	0.062	0.786
combined	0.0	Uncertainty query	0.471	0.821	0.038	0.050	0.726
combined	0.0	Active entropy	0.466	0.818	0.037	0.041	0.732

Table 13: All stress-sweep metrics across six axes, six levels, and seven methods. (continued)

Axis	Level	Method	Success	Eff.	Damage	Unsafe	Utility
combined	0.0	Robust MPC	0.599	1.289	0.034	0.000	0.957
combined	0.0	MIBL v5	0.579	1.146	0.035	0.000	0.925
combined	0.0	Oracle minimal	0.630	1.251	0.036	0.000	0.986
combined	0.2	Full demo	0.477	0.688	0.070	0.088	0.638
combined	0.2	Residual	0.460	0.696	0.045	0.088	0.656
combined	0.2	Uncertainty query	0.452	0.700	0.050	0.081	0.655
combined	0.2	Active entropy	0.428	0.693	0.043	0.062	0.651
combined	0.2	Robust MPC	0.587	1.130	0.034	0.001	0.902
combined	0.2	MIBL v5	0.545	0.973	0.040	0.000	0.843
combined	0.2	Oracle minimal	0.634	1.145	0.037	0.000	0.958
combined	0.4	Full demo	0.400	0.469	0.092	0.128	0.481
combined	0.4	Residual	0.387	0.507	0.055	0.109	0.528
combined	0.4	Uncertainty query	0.459	0.617	0.062	0.112	0.613
combined	0.4	Active entropy	0.465	0.731	0.050	0.081	0.670
combined	0.4	Robust MPC	0.585	0.983	0.036	0.009	0.858
combined	0.4	MIBL v5	0.522	0.845	0.045	0.000	0.770
combined	0.4	Oracle minimal	0.615	0.989	0.042	0.001	0.895
combined	0.6	Full demo	0.322	0.316	0.092	0.153	0.354
combined	0.6	Residual	0.323	0.390	0.053	0.137	0.427
combined	0.6	Uncertainty query	0.384	0.433	0.071	0.145	0.476
combined	0.6	Active entropy	0.422	0.594	0.054	0.097	0.579
combined	0.6	Robust MPC	0.548	0.828	0.044	0.018	0.772
combined	0.6	MIBL v5	0.476	0.696	0.048	0.006	0.680
combined	0.6	Oracle minimal	0.609	0.890	0.046	0.001	0.856
combined	0.8	Full demo	0.257	0.217	0.104	0.174	0.250
combined	0.8	Residual	0.251	0.266	0.048	0.139	0.328
combined	0.8	Uncertainty query	0.305	0.307	0.071	0.161	0.361
combined	0.8	Active entropy	0.367	0.481	0.058	0.113	0.487
combined	0.8	Robust MPC	0.552	0.765	0.055	0.030	0.746
combined	0.8	MIBL v5	0.453	0.605	0.052	0.019	0.624
combined	0.8	Oracle minimal	0.580	0.777	0.046	0.013	0.791
combined	1.0	Full demo	0.197	0.153	0.101	0.207	0.162
combined	1.0	Residual	0.188	0.182	0.056	0.174	0.232
combined	1.0	Uncertainty query	0.278	0.248	0.078	0.189	0.303
combined	1.0	Active entropy	0.339	0.409	0.060	0.129	0.432
combined	1.0	Robust MPC	0.518	0.660	0.061	0.053	0.673
combined	1.0	MIBL v5	0.435	0.532	0.059	0.040	0.574
combined	1.0	Oracle minimal	0.583	0.723	0.055	0.024	0.767

E APPENDIX: COMPLETE FIXED-RISK TABLE

The fixed-risk appendix reports every split, budget, and method. The strict budget 0.05 failure is visible in the full table rather than only in the summary.

Table 14: All fixed-risk deployment rows across both hard splits, four budgets, and six methods.

Split	Budget	Method	Coverage	Succ.	Damage	Unsafe	Utility
Combined	0.02	Active entropy	0.000	0.000	0.000	0.000	0.000
Combined	0.02	MIBL v5	0.000	0.000	0.000	0.000	0.000
Combined	0.02	Oracle minimal	0.000	0.000	0.000	0.000	0.000
Combined	0.02	Robust MPC	0.000	0.000	0.000	0.000	0.000
Combined	0.02	Safety residual	0.000	0.000	0.000	0.000	0.000
Combined	0.02	Uncertainty query	0.000	0.000	0.000	0.000	0.000
Combined	0.05	Active entropy	0.000	0.000	0.000	0.000	0.000
Combined	0.05	MIBL v5	0.000	0.000	0.000	0.000	0.000
Combined	0.05	Oracle minimal	0.000	0.000	0.000	0.000	0.000
Combined	0.05	Robust MPC	0.000	0.000	0.000	0.000	0.000
Combined	0.05	Safety residual	0.000	0.000	0.000	0.000	0.000
Combined	0.05	Uncertainty query	0.000	0.000	0.000	0.000	0.000
Combined	0.08	Active entropy	0.000	0.000	0.000	0.000	0.000
Combined	0.08	MIBL v5	0.000	0.000	0.000	0.000	0.000
Combined	0.08	Oracle minimal	0.090	0.624	0.036	0.000	0.866
Combined	0.08	Robust MPC	0.000	0.000	0.000	0.000	0.000
Combined	0.08	Safety residual	0.000	0.000	0.000	0.000	0.000
Combined	0.08	Uncertainty query	0.000	0.000	0.000	0.000	0.000
Combined	0.10	Active entropy	0.000	0.000	0.000	0.000	0.000
Combined	0.10	MIBL v5	0.000	0.000	0.000	0.000	0.000
Combined	0.10	Oracle minimal	0.997	0.583	0.046	0.008	0.804
Combined	0.10	Robust MPC	0.000	0.000	0.000	0.000	0.000
Combined	0.10	Safety residual	0.000	0.000	0.000	0.000	0.000
Combined	0.10	Uncertainty query	0.000	0.000	0.000	0.000	0.000
Dynamics	0.02	Active entropy	0.000	0.000	0.000	0.000	0.000
Dynamics	0.02	MIBL v5	0.000	0.000	0.000	0.000	0.000
Dynamics	0.02	Oracle minimal	0.000	0.000	0.000	0.000	0.000
Dynamics	0.02	Robust MPC	0.000	0.000	0.000	0.000	0.000
Dynamics	0.02	Safety residual	0.000	0.000	0.000	0.000	0.000
Dynamics	0.02	Uncertainty query	0.000	0.000	0.000	0.000	0.000

Table 14: All fixed-risk deployment rows across both hard splits, four budgets, and six methods. (continued)

Split	Budget	Method	Coverage	Succ.	Damage	Unsafe	Utility
Dynamics	0.05	Active entropy	0.000	0.000	0.000	0.000	0.000
Dynamics	0.05	MIBL v5	0.000	0.000	0.000	0.000	0.000
Dynamics	0.05	Oracle minimal	0.000	0.000	0.000	0.000	0.000
Dynamics	0.05	Robust MPC	0.000	0.000	0.000	0.000	0.000
Dynamics	0.05	Safety residual	0.000	0.000	0.000	0.000	0.000
Dynamics	0.05	Uncertainty query	0.000	0.000	0.000	0.000	0.000
Dynamics	0.08	Active entropy	0.000	0.000	0.000	0.000	0.000
Dynamics	0.08	MIBL v5	0.000	0.000	0.000	0.000	0.000
Dynamics	0.08	Oracle minimal	0.393	0.552	0.034	0.019	0.768
Dynamics	0.08	Robust MPC	0.000	0.000	0.000	0.000	0.000
Dynamics	0.08	Safety residual	0.000	0.000	0.000	0.000	0.000
Dynamics	0.08	Uncertainty query	0.000	0.000	0.000	0.000	0.000
Dynamics	0.10	Active entropy	0.000	0.000	0.000	0.000	0.000
Dynamics	0.10	MIBL v5	0.000	0.000	0.000	0.000	0.000
Dynamics	0.10	Oracle minimal	1.000	0.567	0.039	0.017	0.784
Dynamics	0.10	Robust MPC	0.000	0.000	0.000	0.000	0.000
Dynamics	0.10	Safety residual	0.000	0.000	0.000	0.000	0.000
Dynamics	0.10	Uncertainty query	0.000	0.000	0.000	0.000	0.000

F APPENDIX: FIXED-RISK PAIRED TESTS

The fixed-risk paired table reports coverage and accepted-episode differences across all budgets. It is included because a deployment reviewer would not accept a single budget-only summary.

Table 15: All fixed-risk paired comparisons for MIBL v5 minus each non-v5 reference across budgets.

Split	Budget	Reference	Metric	Mean diff	CI95	Lower95
Combined	0.02	Active entropy	coverage	0.000	0.000	0.000
Combined	0.02	Active entropy	accepted_success	0.000	0.000	0.000
Combined	0.02	Active entropy	accepted_damage	0.000	0.000	0.000
Combined	0.02	Active entropy	accepted_unsafe	0.000	0.000	0.000
Combined	0.02	Active entropy	accepted_utility	0.000	0.000	0.000
Combined	0.02	Oracle minimal	coverage	0.000	0.000	0.000
Combined	0.02	Oracle minimal	accepted_success	0.000	0.000	0.000
Combined	0.02	Oracle minimal	accepted_damage	0.000	0.000	0.000
Combined	0.02	Oracle minimal	accepted_unsafe	0.000	0.000	0.000
Combined	0.02	Oracle minimal	accepted_utility	0.000	0.000	0.000
Combined	0.02	Robust MPC	coverage	0.000	0.000	0.000
Combined	0.02	Robust MPC	accepted_success	0.000	0.000	0.000
Combined	0.02	Robust MPC	accepted_damage	0.000	0.000	0.000
Combined	0.02	Robust MPC	accepted_unsafe	0.000	0.000	0.000
Combined	0.02	Robust MPC	accepted_utility	0.000	0.000	0.000
Combined	0.02	Safety residual	coverage	0.000	0.000	0.000
Combined	0.02	Safety residual	accepted_success	0.000	0.000	0.000
Combined	0.02	Safety residual	accepted_damage	0.000	0.000	0.000
Combined	0.02	Safety residual	accepted_unsafe	0.000	0.000	0.000
Combined	0.02	Safety residual	accepted_utility	0.000	0.000	0.000
Combined	0.02	Uncertainty query	coverage	0.000	0.000	0.000
Combined	0.02	Uncertainty query	accepted_success	0.000	0.000	0.000
Combined	0.02	Uncertainty query	accepted_damage	0.000	0.000	0.000
Combined	0.02	Uncertainty query	accepted_unsafe	0.000	0.000	0.000
Combined	0.02	Uncertainty query	accepted_utility	0.000	0.000	0.000
Combined	0.05	Active entropy	coverage	0.000	0.000	0.000
Combined	0.05	Active entropy	accepted_success	0.000	0.000	0.000
Combined	0.05	Active entropy	accepted_damage	0.000	0.000	0.000
Combined	0.05	Active entropy	accepted_unsafe	0.000	0.000	0.000
Combined	0.05	Active entropy	accepted_utility	0.000	0.000	0.000
Combined	0.05	Oracle minimal	coverage	0.000	0.000	0.000
Combined	0.05	Oracle minimal	accepted_success	0.000	0.000	0.000
Combined	0.05	Oracle minimal	accepted_damage	0.000	0.000	0.000
Combined	0.05	Oracle minimal	accepted_unsafe	0.000	0.000	0.000
Combined	0.05	Oracle minimal	accepted_utility	0.000	0.000	0.000
Combined	0.05	Robust MPC	coverage	0.000	0.000	0.000
Combined	0.05	Robust MPC	accepted_success	0.000	0.000	0.000
Combined	0.05	Robust MPC	accepted_damage	0.000	0.000	0.000
Combined	0.05	Robust MPC	accepted_unsafe	0.000	0.000	0.000
Combined	0.05	Robust MPC	accepted_utility	0.000	0.000	0.000
Combined	0.05	Safety residual	coverage	0.000	0.000	0.000
Combined	0.05	Safety residual	accepted_success	0.000	0.000	0.000
Combined	0.05	Safety residual	accepted_damage	0.000	0.000	0.000
Combined	0.05	Safety residual	accepted_unsafe	0.000	0.000	0.000
Combined	0.05	Safety residual	accepted_utility	0.000	0.000	0.000
Combined	0.05	Uncertainty query	coverage	0.000	0.000	0.000
Combined	0.05	Uncertainty query	accepted_success	0.000	0.000	0.000
Combined	0.05	Uncertainty query	accepted_damage	0.000	0.000	0.000
Combined	0.05	Uncertainty query	accepted_unsafe	0.000	0.000	0.000
Combined	0.05	Uncertainty query	accepted_utility	0.000	0.000	0.000

Table 15: All fixed-risk paired comparisons for MIBL v5 minus each non-v5 reference across budgets. (continued)

Split	Budget	Reference	Metric	Mean diff	CI95	Lower95
Combined	0.08	Active entropy	coverage	0.000	0.000	0.000
Combined	0.08	Active entropy	accepted_success	0.000	0.000	0.000
Combined	0.08	Active entropy	accepted_damage	0.000	0.000	0.000
Combined	0.08	Active entropy	accepted_unsafe	0.000	0.000	0.000
Combined	0.08	Active entropy	accepted_utility	0.000	0.000	0.000
Combined	0.08	Oracle minimal	coverage	-0.090	0.017	-0.106
Combined	0.08	Oracle minimal	accepted_success	-0.624	0.069	-0.693
Combined	0.08	Oracle minimal	accepted_damage	-0.036	0.030	-0.066
Combined	0.08	Oracle minimal	accepted_unsafe	0.000	0.000	0.000
Combined	0.08	Oracle minimal	accepted_utility	-0.866	0.104	-0.971
Combined	0.08	Robust MPC	coverage	0.000	0.000	0.000
Combined	0.08	Robust MPC	accepted_success	0.000	0.000	0.000
Combined	0.08	Robust MPC	accepted_damage	0.000	0.000	0.000
Combined	0.08	Robust MPC	accepted_unsafe	0.000	0.000	0.000
Combined	0.08	Robust MPC	accepted_utility	0.000	0.000	0.000
Combined	0.08	Safety residual	coverage	0.000	0.000	0.000
Combined	0.08	Safety residual	accepted_success	0.000	0.000	0.000
Combined	0.08	Safety residual	accepted_damage	0.000	0.000	0.000
Combined	0.08	Safety residual	accepted_unsafe	0.000	0.000	0.000
Combined	0.08	Safety residual	accepted_utility	0.000	0.000	0.000
Combined	0.08	Uncertainty query	coverage	0.000	0.000	0.000
Combined	0.08	Uncertainty query	accepted_success	0.000	0.000	0.000
Combined	0.08	Uncertainty query	accepted_damage	0.000	0.000	0.000
Combined	0.08	Uncertainty query	accepted_unsafe	0.000	0.000	0.000
Combined	0.08	Uncertainty query	accepted_utility	0.000	0.000	0.000
Combined	0.10	Active entropy	coverage	0.000	0.000	0.000
Combined	0.10	Active entropy	accepted_success	0.000	0.000	0.000
Combined	0.10	Active entropy	accepted_damage	0.000	0.000	0.000
Combined	0.10	Active entropy	accepted_unsafe	0.000	0.000	0.000
Combined	0.10	Active entropy	accepted_utility	0.000	0.000	0.000
Combined	0.10	Oracle minimal	coverage	-0.997	0.002	-0.999
Combined	0.10	Oracle minimal	accepted_success	-0.583	0.020	-0.603
Combined	0.10	Oracle minimal	accepted_damage	-0.046	0.012	-0.058
Combined	0.10	Oracle minimal	accepted_unsafe	-0.008	0.004	-0.013
Combined	0.10	Oracle minimal	accepted_utility	-0.804	0.028	-0.832
Combined	0.10	Robust MPC	coverage	0.000	0.000	0.000
Combined	0.10	Robust MPC	accepted_success	0.000	0.000	0.000
Combined	0.10	Robust MPC	accepted_damage	0.000	0.000	0.000
Combined	0.10	Robust MPC	accepted_unsafe	0.000	0.000	0.000
Combined	0.10	Robust MPC	accepted_utility	0.000	0.000	0.000
Combined	0.10	Safety residual	coverage	0.000	0.000	0.000
Combined	0.10	Safety residual	accepted_success	0.000	0.000	0.000
Combined	0.10	Safety residual	accepted_damage	0.000	0.000	0.000
Combined	0.10	Safety residual	accepted_unsafe	0.000	0.000	0.000
Combined	0.10	Safety residual	accepted_utility	0.000	0.000	0.000
Combined	0.10	Uncertainty query	coverage	0.000	0.000	0.000
Combined	0.10	Uncertainty query	accepted_success	0.000	0.000	0.000
Combined	0.10	Uncertainty query	accepted_damage	0.000	0.000	0.000
Combined	0.10	Uncertainty query	accepted_unsafe	0.000	0.000	0.000
Combined	0.10	Uncertainty query	accepted_utility	0.000	0.000	0.000
Dynamics	0.02	Active entropy	coverage	0.000	0.000	0.000
Dynamics	0.02	Active entropy	accepted_success	0.000	0.000	0.000
Dynamics	0.02	Active entropy	accepted_damage	0.000	0.000	0.000
Dynamics	0.02	Active entropy	accepted_unsafe	0.000	0.000	0.000
Dynamics	0.02	Active entropy	accepted_utility	0.000	0.000	0.000
Dynamics	0.02	Oracle minimal	coverage	0.000	0.000	0.000
Dynamics	0.02	Oracle minimal	accepted_success	0.000	0.000	0.000
Dynamics	0.02	Oracle minimal	accepted_damage	0.000	0.000	0.000
Dynamics	0.02	Oracle minimal	accepted_unsafe	0.000	0.000	0.000
Dynamics	0.02	Oracle minimal	accepted_utility	0.000	0.000	0.000
Dynamics	0.02	Robust MPC	coverage	0.000	0.000	0.000
Dynamics	0.02	Robust MPC	accepted_success	0.000	0.000	0.000
Dynamics	0.02	Robust MPC	accepted_damage	0.000	0.000	0.000
Dynamics	0.02	Robust MPC	accepted_unsafe	0.000	0.000	0.000
Dynamics	0.02	Robust MPC	accepted_utility	0.000	0.000	0.000
Dynamics	0.02	Safety residual	coverage	0.000	0.000	0.000
Dynamics	0.02	Safety residual	accepted_success	0.000	0.000	0.000
Dynamics	0.02	Safety residual	accepted_damage	0.000	0.000	0.000
Dynamics	0.02	Safety residual	accepted_unsafe	0.000	0.000	0.000
Dynamics	0.02	Safety residual	accepted_utility	0.000	0.000	0.000
Dynamics	0.02	Uncertainty query	coverage	0.000	0.000	0.000
Dynamics	0.02	Uncertainty query	accepted_success	0.000	0.000	0.000
Dynamics	0.02	Uncertainty query	accepted_damage	0.000	0.000	0.000
Dynamics	0.02	Uncertainty query	accepted_unsafe	0.000	0.000	0.000
Dynamics	0.02	Uncertainty query	accepted_utility	0.000	0.000	0.000
Dynamics	0.05	Active entropy	coverage	0.000	0.000	0.000
Dynamics	0.05	Active entropy	accepted_success	0.000	0.000	0.000
Dynamics	0.05	Active entropy	accepted_damage	0.000	0.000	0.000
Dynamics	0.05	Active entropy	accepted_unsafe	0.000	0.000	0.000
Dynamics	0.05	Active entropy	accepted_utility	0.000	0.000	0.000
Dynamics	0.05	Oracle minimal	coverage	0.000	0.000	0.000
Dynamics	0.05	Oracle minimal	accepted_success	0.000	0.000	0.000
Dynamics	0.05	Oracle minimal	accepted_damage	0.000	0.000	0.000
Dynamics	0.05	Oracle minimal	accepted_unsafe	0.000	0.000	0.000
Dynamics	0.05	Oracle minimal	accepted_utility	0.000	0.000	0.000
Dynamics	0.05	Robust MPC	coverage	0.000	0.000	0.000
Dynamics	0.05	Robust MPC	accepted_success	0.000	0.000	0.000
Dynamics	0.05	Robust MPC	accepted_damage	0.000	0.000	0.000
Dynamics	0.05	Robust MPC	accepted_unsafe	0.000	0.000	0.000

Table 15: All fixed-risk paired comparisons for MIBL v5 minus each non-v5 reference across budgets. (continued)

Split	Budget	Reference	Metric	Mean diff	CI95	Lower95
Dynamics	0.05	Robust MPC	accepted_utility	0.000	0.000	0.000
Dynamics	0.05	Safety residual	coverage	0.000	0.000	0.000
Dynamics	0.05	Safety residual	accepted_success	0.000	0.000	0.000
Dynamics	0.05	Safety residual	accepted_damage	0.000	0.000	0.000
Dynamics	0.05	Safety residual	accepted_unsafe	0.000	0.000	0.000
Dynamics	0.05	Safety residual	accepted_utility	0.000	0.000	0.000
Dynamics	0.05	Uncertainty query	coverage	0.000	0.000	0.000
Dynamics	0.05	Uncertainty query	accepted_success	0.000	0.000	0.000
Dynamics	0.05	Uncertainty query	accepted_damage	0.000	0.000	0.000
Dynamics	0.05	Uncertainty query	accepted_unsafe	0.000	0.000	0.000
Dynamics	0.05	Uncertainty query	accepted_utility	0.000	0.000	0.000
Dynamics	0.08	Active entropy	coverage	0.000	0.000	0.000
Dynamics	0.08	Active entropy	accepted_success	0.000	0.000	0.000
Dynamics	0.08	Active entropy	accepted_damage	0.000	0.000	0.000
Dynamics	0.08	Active entropy	accepted_unsafe	0.000	0.000	0.000
Dynamics	0.08	Active entropy	accepted_utility	0.000	0.000	0.000
Dynamics	0.08	Oracle minimal	coverage	-0.393	0.020	-0.413
Dynamics	0.08	Oracle minimal	accepted_success	-0.552	0.042	-0.594
Dynamics	0.08	Oracle minimal	accepted_damage	-0.034	0.010	-0.045
Dynamics	0.08	Oracle minimal	accepted_unsafe	-0.019	0.008	-0.027
Dynamics	0.08	Oracle minimal	accepted_utility	-0.768	0.055	-0.823
Dynamics	0.08	Robust MPC	coverage	0.000	0.000	0.000
Dynamics	0.08	Robust MPC	accepted_success	0.000	0.000	0.000
Dynamics	0.08	Robust MPC	accepted_damage	0.000	0.000	0.000
Dynamics	0.08	Robust MPC	accepted_unsafe	0.000	0.000	0.000
Dynamics	0.08	Robust MPC	accepted_utility	0.000	0.000	0.000
Dynamics	0.08	Safety residual	coverage	0.000	0.000	0.000
Dynamics	0.08	Safety residual	accepted_success	0.000	0.000	0.000
Dynamics	0.08	Safety residual	accepted_damage	0.000	0.000	0.000
Dynamics	0.08	Safety residual	accepted_unsafe	0.000	0.000	0.000
Dynamics	0.08	Safety residual	accepted_utility	0.000	0.000	0.000
Dynamics	0.08	Uncertainty query	coverage	0.000	0.000	0.000
Dynamics	0.08	Uncertainty query	accepted_success	0.000	0.000	0.000
Dynamics	0.08	Uncertainty query	accepted_damage	0.000	0.000	0.000
Dynamics	0.08	Uncertainty query	accepted_unsafe	0.000	0.000	0.000
Dynamics	0.08	Uncertainty query	accepted_utility	0.000	0.000	0.000
Dynamics	0.10	Active entropy	coverage	0.000	0.000	0.000
Dynamics	0.10	Active entropy	accepted_success	0.000	0.000	0.000
Dynamics	0.10	Active entropy	accepted_damage	0.000	0.000	0.000
Dynamics	0.10	Active entropy	accepted_unsafe	0.000	0.000	0.000
Dynamics	0.10	Active entropy	accepted_utility	0.000	0.000	0.000
Dynamics	0.10	Oracle minimal	coverage	-1.000	0.000	-1.000
Dynamics	0.10	Oracle minimal	accepted_success	-0.567	0.014	-0.581
Dynamics	0.10	Oracle minimal	accepted_damage	-0.039	0.005	-0.044
Dynamics	0.10	Oracle minimal	accepted_unsafe	-0.017	0.005	-0.021
Dynamics	0.10	Oracle minimal	accepted_utility	-0.784	0.019	-0.803
Dynamics	0.10	Robust MPC	coverage	0.000	0.000	0.000
Dynamics	0.10	Robust MPC	accepted_success	0.000	0.000	0.000
Dynamics	0.10	Robust MPC	accepted_damage	0.000	0.000	0.000
Dynamics	0.10	Robust MPC	accepted_unsafe	0.000	0.000	0.000
Dynamics	0.10	Robust MPC	accepted_utility	0.000	0.000	0.000
Dynamics	0.10	Safety residual	coverage	0.000	0.000	0.000
Dynamics	0.10	Safety residual	accepted_success	0.000	0.000	0.000
Dynamics	0.10	Safety residual	accepted_damage	0.000	0.000	0.000
Dynamics	0.10	Safety residual	accepted_unsafe	0.000	0.000	0.000
Dynamics	0.10	Safety residual	accepted_utility	0.000	0.000	0.000
Dynamics	0.10	Uncertainty query	coverage	0.000	0.000	0.000
Dynamics	0.10	Uncertainty query	accepted_success	0.000	0.000	0.000
Dynamics	0.10	Uncertainty query	accepted_damage	0.000	0.000	0.000
Dynamics	0.10	Uncertainty query	accepted_unsafe	0.000	0.000	0.000
Dynamics	0.10	Uncertainty query	accepted_utility	0.000	0.000	0.000

G APPENDIX: ABLATION SEED UNITS

The mechanism gate was evaluated on seed means from dynamics mismatch and combined hard shift. The full mechanism does not achieve the required robust-utility margin over all ablations.

Table 16: Ablation seed means for the two hard ablation splits.

Split	Ablation	Seed	Success	Efficiency	Damage	Utility
Combined	all_corrections_imitation	0	0.363	0.367	0.060	0.482
Combined	all_corrections_imitation	1	0.405	0.415	0.071	0.516
Combined	all_corrections_imitation	2	0.345	0.348	0.060	0.451
Combined	all_corrections_imitation	3	0.321	0.322	0.065	0.419
Combined	all_corrections_imitation	4	0.310	0.339	0.048	0.426
Combined	all_corrections_imitation	5	0.292	0.304	0.071	0.385
Combined	all_corrections_imitation	6	0.351	0.358	0.042	0.468

Table 16: Ablation seed means for the two hard ablation splits. (continued)

Split	Ablation	Seed	Success	Efficiency	Damage	Utility
Combined	all_corrections_imitation	7	0.327	0.348	0.060	0.439
Combined	all_corrections_imitation	8	0.321	0.338	0.054	0.434
Combined	all_corrections_imitation	9	0.363	0.405	0.048	0.492
Combined	full_minimum_intervention_boundary_learner.v5	0	0.500	0.666	0.054	0.691
Combined	full_minimum_intervention_boundary_learner.v5	1	0.440	0.571	0.030	0.619
Combined	full_minimum_intervention_boundary_learner.v5	2	0.554	0.747	0.042	0.770
Combined	full_minimum_intervention_boundary_learner.v5	3	0.488	0.665	0.065	0.674
Combined	full_minimum_intervention_boundary_learner.v5	4	0.458	0.654	0.054	0.643
Combined	full_minimum_intervention_boundary_learner.v5	5	0.429	0.565	0.065	0.590
Combined	full_minimum_intervention_boundary_learner.v5	6	0.464	0.622	0.024	0.660
Combined	full_minimum_intervention_boundary_learner.v5	7	0.488	0.673	0.042	0.682
Combined	full_minimum_intervention_boundary_learner.v5	8	0.464	0.674	0.060	0.657
Combined	full_minimum_intervention_boundary_learner.v5	9	0.470	0.648	0.048	0.658
Combined	minus_calibration	0	0.607	0.823	0.065	0.828
Combined	minus_calibration	1	0.446	0.606	0.054	0.622
Combined	minus_calibration	2	0.536	0.727	0.018	0.757
Combined	minus_calibration	3	0.429	0.586	0.024	0.613
Combined	minus_calibration	4	0.440	0.610	0.060	0.614
Combined	minus_calibration	5	0.494	0.670	0.048	0.685
Combined	minus_calibration	6	0.518	0.726	0.042	0.725
Combined	minus_calibration	7	0.494	0.689	0.054	0.690
Combined	minus_calibration	8	0.506	0.697	0.048	0.702
Combined	minus_calibration	9	0.500	0.670	0.060	0.685
Combined	minus_counterfactual_boundary	0	0.351	0.457	0.030	0.516
Combined	minus_counterfactual_boundary	1	0.488	0.598	0.077	0.658
Combined	minus_counterfactual_boundary	2	0.506	0.658	0.024	0.717
Combined	minus_counterfactual_boundary	3	0.452	0.612	0.012	0.660
Combined	minus_counterfactual_boundary	4	0.435	0.557	0.042	0.612
Combined	minus_counterfactual_boundary	5	0.375	0.473	0.060	0.524
Combined	minus_counterfactual_boundary	6	0.399	0.527	0.048	0.560
Combined	minus_counterfactual_boundary	7	0.458	0.602	0.030	0.651
Combined	minus_counterfactual_boundary	8	0.429	0.591	0.042	0.618
Combined	minus_counterfactual_boundary	9	0.458	0.594	0.048	0.641
Combined	minus_human_effort_cost	0	0.542	0.748	0.077	0.732
Combined	minus_human_effort_cost	1	0.542	0.727	0.060	0.734
Combined	minus_human_effort_cost	2	0.607	0.837	0.065	0.825
Combined	minus_human_effort_cost	3	0.488	0.704	0.048	0.676
Combined	minus_human_effort_cost	4	0.601	0.838	0.065	0.817
Combined	minus_human_effort_cost	5	0.518	0.720	0.048	0.721
Combined	minus_human_effort_cost	6	0.524	0.742	0.030	0.738
Combined	minus_human_effort_cost	7	0.595	0.867	0.054	0.825
Combined	minus_human_effort_cost	8	0.554	0.815	0.042	0.778
Combined	minus_human_effort_cost	9	0.536	0.744	0.060	0.731
Combined	minus_intent_preservation	0	0.393	0.514	0.065	0.545
Combined	minus_intent_preservation	1	0.405	0.549	0.024	0.584
Combined	minus_intent_preservation	2	0.488	0.669	0.054	0.674
Combined	minus_intent_preservation	3	0.548	0.761	0.065	0.757
Combined	minus_intent_preservation	4	0.488	0.662	0.036	0.686
Combined	minus_intent_preservation	5	0.470	0.633	0.024	0.670
Combined	minus_intent_preservation	6	0.518	0.707	0.054	0.711
Combined	minus_intent_preservation	7	0.399	0.556	0.071	0.551
Combined	minus_intent_preservation	8	0.482	0.682	0.036	0.687
Combined	minus_intent_preservation	9	0.512	0.680	0.048	0.708
Combined	minus_minimum_norm_objective	0	0.494	0.679	0.071	0.667
Combined	minus_minimum_norm_objective	1	0.548	0.750	0.048	0.752
Combined	minus_minimum_norm_objective	2	0.542	0.746	0.024	0.758
Combined	minus_minimum_norm_objective	3	0.506	0.712	0.036	0.708
Combined	minus_minimum_norm_objective	4	0.488	0.664	0.065	0.661
Combined	minus_minimum_norm_objective	5	0.583	0.794	0.042	0.802
Combined	minus_minimum_norm_objective	6	0.560	0.790	0.054	0.762
Combined	minus_minimum_norm_objective	7	0.607	0.871	0.119	0.800
Combined	minus_minimum_norm_objective	8	0.530	0.756	0.036	0.746
Combined	minus_minimum_norm_objective	9	0.577	0.794	0.060	0.785
Combined	minus_query_throttling	0	0.524	0.704	0.018	0.742
Combined	minus_query_throttling	1	0.423	0.557	0.048	0.589
Combined	minus_query_throttling	2	0.488	0.652	0.036	0.677
Combined	minus_query_throttling	3	0.542	0.718	0.024	0.760
Combined	minus_query_throttling	4	0.500	0.668	0.048	0.692
Combined	minus_query_throttling	5	0.518	0.681	0.060	0.712
Combined	minus_query_throttling	6	0.488	0.676	0.024	0.692
Combined	minus_query_throttling	7	0.500	0.687	0.054	0.694
Combined	minus_query_throttling	8	0.488	0.686	0.036	0.688
Combined	minus_query_throttling	9	0.494	0.689	0.060	0.685
Combined	minus_safety_override	0	0.542	0.789	0.054	0.742
Combined	minus_safety_override	1	0.482	0.652	0.060	0.657
Combined	minus_safety_override	2	0.589	0.832	0.048	0.805
Combined	minus_safety_override	3	0.536	0.743	0.036	0.746
Combined	minus_safety_override	4	0.488	0.678	0.036	0.679
Combined	minus_safety_override	5	0.512	0.702	0.065	0.686
Combined	minus_safety_override	6	0.583	0.835	0.054	0.796
Combined	minus_safety_override	7	0.560	0.833	0.036	0.776
Combined	minus_safety_override	8	0.542	0.785	0.060	0.747
Combined	minus_safety_override	9	0.494	0.709	0.065	0.668
Combined	preference_only_objective	0	0.411	0.546	0.065	0.563
Combined	preference_only_objective	1	0.458	0.601	0.077	0.619
Combined	preference_only_objective	2	0.435	0.595	0.065	0.599
Combined	preference_only_objective	3	0.482	0.646	0.054	0.662
Combined	preference_only_objective	4	0.440	0.590	0.065	0.605
Combined	preference_only_objective	5	0.435	0.566	0.071	0.595

Table 16: Ablation seed means for the two hard ablation splits. (continued)

Split	Ablation	Seed	Success	Efficiency	Damage	Utility
Combined	preference_only_objective	6	0.464	0.641	0.089	0.629
Combined	preference_only_objective	7	0.476	0.647	0.042	0.667
Combined	preference_only_objective	8	0.446	0.629	0.054	0.624
Combined	preference_only_objective	9	0.435	0.583	0.089	0.572
Dynamics	all_corrections_imitation	0	0.476	0.546	0.042	0.639
Dynamics	all_corrections_imitation	1	0.250	0.302	0.030	0.364
Dynamics	all_corrections_imitation	2	0.470	0.537	0.071	0.612
Dynamics	all_corrections_imitation	3	0.411	0.468	0.036	0.566
Dynamics	all_corrections_imitation	4	0.429	0.502	0.054	0.587
Dynamics	all_corrections_imitation	5	0.458	0.524	0.071	0.604
Dynamics	all_corrections_imitation	6	0.494	0.556	0.101	0.633
Dynamics	all_corrections_imitation	7	0.393	0.450	0.042	0.543
Dynamics	all_corrections_imitation	8	0.363	0.416	0.054	0.499
Dynamics	all_corrections_imitation	9	0.429	0.509	0.030	0.596
Dynamics	full_minimum_intervention_boundary_learner.v5	0	0.482	0.629	0.054	0.668
Dynamics	full_minimum_intervention_boundary_learner.v5	1	0.500	0.680	0.071	0.688
Dynamics	full_minimum_intervention_boundary_learner.v5	2	0.494	0.666	0.077	0.672
Dynamics	full_minimum_intervention_boundary_learner.v5	3	0.470	0.624	0.030	0.658
Dynamics	full_minimum_intervention_boundary_learner.v5	4	0.464	0.641	0.042	0.663
Dynamics	full_minimum_intervention_boundary_learner.v5	5	0.452	0.585	0.077	0.619
Dynamics	full_minimum_intervention_boundary_learner.v5	6	0.518	0.674	0.024	0.733
Dynamics	full_minimum_intervention_boundary_learner.v5	7	0.512	0.677	0.024	0.714
Dynamics	full_minimum_intervention_boundary_learner.v5	8	0.571	0.742	0.048	0.782
Dynamics	full_minimum_intervention_boundary_learner.v5	9	0.440	0.578	0.077	0.599
Dynamics	minus_calibration	0	0.530	0.683	0.060	0.724
Dynamics	minus_calibration	1	0.512	0.686	0.083	0.693
Dynamics	minus_calibration	2	0.530	0.695	0.036	0.737
Dynamics	minus_calibration	3	0.518	0.666	0.018	0.724
Dynamics	minus_calibration	4	0.595	0.808	0.030	0.827
Dynamics	minus_calibration	5	0.542	0.729	0.036	0.760
Dynamics	minus_calibration	6	0.488	0.657	0.077	0.672
Dynamics	minus_calibration	7	0.560	0.759	0.048	0.779
Dynamics	minus_calibration	8	0.440	0.590	0.018	0.632
Dynamics	minus_calibration	9	0.435	0.578	0.042	0.612
Dynamics	minus_counterfactual_boundary	0	0.458	0.590	0.036	0.644
Dynamics	minus_counterfactual_boundary	1	0.452	0.586	0.024	0.646
Dynamics	minus_counterfactual_boundary	2	0.458	0.567	0.042	0.642
Dynamics	minus_counterfactual_boundary	3	0.482	0.617	0.024	0.685
Dynamics	minus_counterfactual_boundary	4	0.440	0.570	0.030	0.634
Dynamics	minus_counterfactual_boundary	5	0.357	0.440	0.042	0.512
Dynamics	minus_counterfactual_boundary	6	0.429	0.561	0.048	0.612
Dynamics	minus_counterfactual_boundary	7	0.381	0.495	0.065	0.531
Dynamics	minus_counterfactual_boundary	8	0.411	0.523	0.042	0.587
Dynamics	minus_counterfactual_boundary	9	0.440	0.575	0.054	0.613
Dynamics	minus_human_effort_cost	0	0.595	0.777	0.077	0.790
Dynamics	minus_human_effort_cost	1	0.613	0.837	0.071	0.825
Dynamics	minus_human_effort_cost	2	0.583	0.792	0.065	0.790
Dynamics	minus_human_effort_cost	3	0.619	0.832	0.018	0.859
Dynamics	minus_human_effort_cost	4	0.554	0.762	0.054	0.756
Dynamics	minus_human_effort_cost	5	0.548	0.728	0.024	0.760
Dynamics	minus_human_effort_cost	6	0.577	0.788	0.030	0.807
Dynamics	minus_human_effort_cost	7	0.577	0.792	0.030	0.803
Dynamics	minus_human_effort_cost	8	0.607	0.827	0.048	0.835
Dynamics	minus_human_effort_cost	9	0.530	0.717	0.048	0.726
Dynamics	minus_intent_preservation	0	0.476	0.617	0.030	0.667
Dynamics	minus_intent_preservation	1	0.530	0.694	0.036	0.737
Dynamics	minus_intent_preservation	2	0.524	0.689	0.060	0.712
Dynamics	minus_intent_preservation	3	0.554	0.720	0.060	0.748
Dynamics	minus_intent_preservation	4	0.435	0.576	0.048	0.608
Dynamics	minus_intent_preservation	5	0.488	0.644	0.048	0.678
Dynamics	minus_intent_preservation	6	0.506	0.698	0.054	0.701
Dynamics	minus_intent_preservation	7	0.435	0.574	0.071	0.593
Dynamics	minus_intent_preservation	8	0.530	0.692	0.036	0.742
Dynamics	minus_intent_preservation	9	0.446	0.605	0.042	0.630
Dynamics	minus_minimum_norm_objective	0	0.571	0.763	0.060	0.778
Dynamics	minus_minimum_norm_objective	1	0.565	0.772	0.036	0.788
Dynamics	minus_minimum_norm_objective	2	0.637	0.852	0.065	0.856
Dynamics	minus_minimum_norm_objective	3	0.571	0.785	0.024	0.798
Dynamics	minus_minimum_norm_objective	4	0.560	0.782	0.042	0.785
Dynamics	minus_minimum_norm_objective	5	0.512	0.676	0.042	0.700
Dynamics	minus_minimum_norm_objective	6	0.565	0.763	0.042	0.775
Dynamics	minus_minimum_norm_objective	7	0.625	0.830	0.054	0.849
Dynamics	minus_minimum_norm_objective	8	0.565	0.748	0.048	0.772
Dynamics	minus_minimum_norm_objective	9	0.577	0.770	0.077	0.781
Dynamics	minus_query_throttling	0	0.524	0.683	0.054	0.721
Dynamics	minus_query_throttling	1	0.488	0.664	0.048	0.687
Dynamics	minus_query_throttling	2	0.577	0.749	0.036	0.793
Dynamics	minus_query_throttling	3	0.542	0.717	0.018	0.761
Dynamics	minus_query_throttling	4	0.542	0.738	0.036	0.766
Dynamics	minus_query_throttling	5	0.560	0.747	0.012	0.788
Dynamics	minus_query_throttling	6	0.530	0.712	0.048	0.734
Dynamics	minus_query_throttling	7	0.548	0.735	0.030	0.765
Dynamics	minus_query_throttling	8	0.488	0.648	0.071	0.666
Dynamics	minus_query_throttling	9	0.536	0.719	0.024	0.758
Dynamics	minus_safety_override	0	0.637	0.852	0.077	0.839
Dynamics	minus_safety_override	1	0.565	0.772	0.065	0.767
Dynamics	minus_safety_override	2	0.607	0.817	0.060	0.811
Dynamics	minus_safety_override	3	0.494	0.660	0.083	0.665
Dynamics	minus_safety_override	4	0.589	0.810	0.065	0.800

Table 16: Ablation seed means for the two hard ablation splits. (continued)

Split	Ablation	Seed	Success	Efficiency	Damage	Utility
Dynamics	minus_safety_override	5	0.595	0.809	0.024	0.817
Dynamics	minus_safety_override	6	0.542	0.746	0.060	0.739
Dynamics	minus_safety_override	7	0.565	0.782	0.030	0.783
Dynamics	minus_safety_override	8	0.613	0.820	0.065	0.819
Dynamics	minus_safety_override	9	0.583	0.795	0.054	0.793
Dynamics	preference_only_objective	0	0.470	0.617	0.065	0.654
Dynamics	preference_only_objective	1	0.440	0.588	0.060	0.613
Dynamics	preference_only_objective	2	0.506	0.654	0.042	0.698
Dynamics	preference_only_objective	3	0.393	0.511	0.065	0.537
Dynamics	preference_only_objective	4	0.554	0.745	0.083	0.753
Dynamics	preference_only_objective	5	0.464	0.590	0.036	0.653
Dynamics	preference_only_objective	6	0.548	0.727	0.071	0.745
Dynamics	preference_only_objective	7	0.470	0.616	0.077	0.639
Dynamics	preference_only_objective	8	0.482	0.632	0.030	0.678
Dynamics	preference_only_objective	9	0.417	0.551	0.036	0.585

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